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MSci Research Project

Simultaneous Dimensionality Reduction and Clustering Methods with Disjointness

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Submitted in partial fulfillment of the requirements for the MSci in Mathematics at Imperial College London

June 2026

Abstract

Simultaneous dimensionality reduction and clustering methods aim to identify a reduced subspace configured to reveal an underlying cluster structure for objects that are described by a set of high-dimensional features. While effective at mitigating the effects of masking variables, existing procedures produce dense loading matrices that can obscure individual feature contributions and hinder interpretability. To resolve this, several authors have introduced disjointness constraints to simpler models, aiming to produce latent components which are each linear combinations of a subset of features, where all the subsets form a partition of features. However, these constrained methods remain vulnerable when irrelevant variables exhibit correlation among themselves.

Drawing influence from these existing approaches, a novel methodology termed Disjoint Generalised Reduced Clustering (DGRC) is proposed. This model extends the concept of disjoint variable partitioning onto the Generalised Reduced Clustering (GRC) framework, ensuring that each original feature contributes to exactly one latent component. The objective is optimised using an original alternating least-squares algorithm. This iterative procedure is initially developed for a simpler framework with the aim of bypassing computationally expensive eigenvalue decompositions, and is thereafter extended to DGRC. The ability of the proposed methodology in recovering accurate cluster partitions is evaluated against preceding baseline methods through controlled synthetic simulations and a real-world dataset.

Acknowledgements

I would like to express my gratitude and appreciation towards my supervisor, *Dr. Efthymios Costa* for his continued support and guidance throughout the completion of this project, and particularly the detailed feedback which I had received that was essential in refining and improving my work. I would also like to thank the various members of staff who have supported me over the last four years at Imperial, as they have each played a part in helping me reach the end of my MSci degree. I have truly gained an incredible amount of knowledge and benefited from invaluable experiences throughout my time here.

Plagiarism statement

The work contained in this thesis is my own work unless otherwise stated.

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Date: June 9, 2026

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Chapter 1

Introduction

High-dimensional data appears across a wide range of academic and industrial fields, such as computational biology, natural language processing, and acoustic signal processing, where individual observations are characterised by dozens to thousands of features. As dimensionality scales up in such regimes, standard clustering procedures such as K-means can often fail to obtain informative cluster groupings, as distance metrics lose their meaningfulness, a phenomenon widely known as the curse of dimensionality Bellman (1961). This necessitates the application of dimensionality reduction techniques prior to clustering. The most common approach involves using Principal Component Analysis (PCA) to reduce the feature space, followed by applying K-means in the reduced feature space. However, this sequential procedure termed 'tandem analysis' by Arabie and Hubert (1994), possesses a fundamental vulnerability in this context. There is no theoretical guarantee that the dimensions capturing maximal variance, which are obtained by PCA, are relevant to an underlying cluster structure. In the presence of high variance noise, PCA will preferentially retain such masking variables which can therefore distort or completely discard the true taxonomic information.

To address the vulnerabilities of tandem analysis, De Soete and Carroll (1994) introduce Reduced K-means (RKM), proposing a single objective function that couples dimensionality reduction and clustering. RKM prioritises cluster separation over total variance and introduces an alternating least-squares scheme which alternates between updating a low-dimensional projection matrix and the cluster assignments, ensuring both steps inform one another. However, Vichi and Kiers (2001) note that RKM can still struggle when the original space contains substantial variance orthogonal to the clustering structure, and propose Factorial K-means (FKM) to optimise within-cluster distances directly within the projected subspace. While RKM and FKM provide ways to obtain subspaces which are relevant to a cluster structure, they implicitly assume that the residual variance is entirely uncorrelated. Recognising that irrelevant or disturbing variables can often exhibit correlations between themselves, Yamamoto and Hwang (2014) introduce Generalised Reduced Clustering (GRC). GRC proposes splitting the reduced subspace further into two orthogonal subspaces, which isolate variables relevant to the cluster structure from the disturbing variables. While effective at reducing dimensionality, these methods rely on eigenvector decompositions which distribute loading weights across all variables. This can lead to a lack of interpretability within the latent space and makes it difficult to attribute cluster assignments to specific features within the original high-dimensional space.
To resolve this lack of interpretability, Vichi (2002) and Vichi and Saporta (2009) introduce Disjoint Factorial K-means (DFKM) and Clustering and Disjoint Principal Component Analysis (CDPCA) which extend FKM and RKM respectively by imposing a disjointness constraint on the loading matrix, ensuring that each original feature contributes to exactly one latent component. Vichi and Saporta (2009) propose a method to optimise its objective using a similar alternating least-squares procedure, whereas Vichi (2002) rely on sequential quadratic programming for DFKM.

This project first bridges this algorithmic gap by developing an original alternating least-squares loop for DFKM that avoids computationally expensive decompositions. Utilising this loop, we present the primary methodological contribution of this report by extending the notion of disjointness onto the GRC framework to formulate a novel Disjoint Generalised Reduced Clustering (DGRC) procedure. Furthermore, we address the resulting vulnerabilities concerning low-variance subspaces and detail various approaches to circumvent these issues. Finally, we evaluate the performance of DGRC against preceding methods using controlled synthetic simulations and a real-world dataset consisting of Global Health Indicators.

The remainder of this report is structured as follows. Chapter 2 establishes the preliminary mathematical notation and demonstrates the fundamental flaws of tandem analysis. Chapter 3 details the theoretical foundations of simultaneous dimensionality reduction and clustering models, specifically covering RKM, FKM, and GRC. Chapter 4 formalises the notion of disjointness constraints within the CDPCA, DFKM and DGRC frameworks. Chapter 5 presents the evaluation of these methodologies through a synthetic simulation study and an application to global health data. Chapter 6 concludes the project by summarising the core findings and identifying potential avenues for future exploration.

Chapter 2

Preliminaries

2.1 Notation

N, P	Number of objects and number of variables
K	Number of clusters where the number of points in each cluster is denoted by $\{n_1, n_2, \ldots, n_K\}$.
L	Number of factors
$\mathbf{X} = (x_{np})$	$N \times P$ data matrix with centered columns, i.e. $\sum_{n=1}^{N} x_{np} = 0$
$\mathbf{U} = (u_{nk})$	$N \times K$ binary row-stochastic cluster membership matrix with $\text{rank}(\mathbf{U}) = K$, where $u_{nk} = 1$ if object n belongs to cluster k, and 0 otherwise. Row-stochastic property means each object belongs to exactly one cluster.
$\mathbf{A} = (a_{pl})$	$P \times L$ columnwise-orthonormal loading matrix, $\mathbf{A}^\top \mathbf{A} = \mathbf{I}_L$
$\mathbf{P_U}$	$N \times N$ orthogonal projection matrix onto the column space of $\mathbf{U}$. Explicitly defined as $\mathbf{U}(\mathbf{U}^\top\mathbf{U})^{-1}\mathbf{U}^\top$.
$\lVert \cdot \rVert_F$	Frobenius norm
$\text{Tr}(\cdot)$	Matrix trace operator
$\odot$	Hadamard element-wise matrix product

When the reduced space represented by $\mathbf{A}$ is partitioned further, we consider additionally with $L = L_C + L_D$ and $\mathbf{A} = (\mathbf{A}_C | \mathbf{A}_D)$:

L_C, L_D	Number of variables relevant to clustering and number of disturbing variables
$\mathbf{A}_C$	$P \times L_C$ columnwise-orthonormal loading matrix that projects onto subspace of clustering relevant variables
$\mathbf{A}_D$	$P \times L_D$ columnwise-orthonormal loading matrix that projects onto subspace of disturbing variables

In the formulation proposed by Vichi and Kiers (2001), the authors introduce an $N \times L$ matrix of factor scores, $\mathbf{Y} = \mathbf{X}\mathbf{A}$, which represents the projection of the original objects onto the reduced L-dimensional space.
Letting $\bar{\mathbf{Y}}$ be the $K \times L$ matrix of K cluster centroids in the reduced space we can derive,

$$\bar{\mathbf{Y}} = (\mathbf{U}^\top\mathbf{U})^{-1}\mathbf{U}^\top\mathbf{Y}$$

Note that $\mathbf{U}^\top\mathbf{U}$ is the diagonal matrix $\text{diag}(n_1, n_2, \ldots, n_K)$, so $(\mathbf{U}^\top\mathbf{U})^{-1}$ scales each cluster by its cluster mass to obtain cluster centroids.

To assign each of the N objects to their corresponding centroids we can left-multiply by $\mathbf{U}$:

$$\mathbf{U}\bar{\mathbf{Y}} = \mathbf{U}(\mathbf{U}^\top\mathbf{U})^{-1}\mathbf{U}^\top\mathbf{Y}$$

The term $\mathbf{U}(\mathbf{U}^\top\mathbf{U})^{-1}\mathbf{U}^\top$ on the RHS is exactly what we have defined for $\mathbf{P}_\mathbf{U}$. In other words, this means that all $\mathbf{P}_\mathbf{U}$ does is project a set of N objects to their corresponding cluster centroids based on some cluster membership $\mathbf{U}$.

Although the explicit calculation of $\mathbf{Y}$ and $\bar{\mathbf{Y}}$ is central to the algorithmic steps in Vichi and Kiers (2001), geometrically they represent intermediate stages of the projection. Therefore from here on, we omit the $\mathbf{Y}$ and $\bar{\mathbf{Y}}$ and formulate the model and objective functions directly using the projection matrix $\mathbf{P}_\mathbf{U}$ acting on $\mathbf{XA}$, as done by Yamamoto and Hwang (2014).

2.2 K-means

K-means clustering MacQueen (1967) is a procedure for partitioning N objects into K disjoint groups such that the sum of squared distances between points and their corresponding cluster centroids is minimised. Formally, it aims to find the optimal cluster parition $\mathbf{U}$ that satisfies:

$$\underset{\mathbf{U}}{\operatorname{argmin}} \sum_{n=1}^{N}\sum_{k=1}^{K} \mathbf{U}_{n,k}\|\mathbf{X}_{n,\bullet} - \bar{\mathbf{X}}_{k,\bullet}\|_2^2$$

where the subscripts represent matrix elements while $\mathbf{X}_{n,\bullet}$ represents the n^{th} row of $\mathbf{X}$, and $\bar{\mathbf{X}}$ represents the cluster centroids. This objective can be rewritten in the following equivalent form:

$$\ell_{KM}(\mathbf{U}) = \|\mathbf{X} - \mathbf{P}_\mathbf{U}\mathbf{X}\|_F^2 \tag{2.1}$$

where $\mathbf{P}_\mathbf{U}$ is the orthogonal projection matrix onto the cluster subspace. Using the properties of the Frobenius norm and the idempotence of $\mathbf{P}_\mathbf{U}$, the objective can be expanded as:

$$\begin{aligned}
\|\mathbf{X} - \mathbf{P}_\mathbf{U}\mathbf{X}\|_F^2 &= \operatorname{Tr}\left((\mathbf{X} - \mathbf{P}_\mathbf{U}\mathbf{X})^\top(\mathbf{X} - \mathbf{P}_\mathbf{U}\mathbf{X})\right) \\
&= \operatorname{Tr}\left(\mathbf{X}^\top\mathbf{X} - \mathbf{X}^\top\mathbf{P}_\mathbf{U}\mathbf{X} - \mathbf{X}^\top\mathbf{P}_\mathbf{U}^\top\mathbf{X} + \mathbf{X}^\top\mathbf{P}_\mathbf{U}^\top\mathbf{P}_\mathbf{U}\mathbf{X}\right) \\
&= \operatorname{Tr}(\mathbf{X}^\top\mathbf{X}) - \operatorname{Tr}(\mathbf{X}^\top\mathbf{P}_\mathbf{U}\mathbf{X})
\end{aligned}$$

Since the total variance $\operatorname{Tr}(\mathbf{X}^\top\mathbf{X})$ is constant with respect to the partition $\mathbf{U}$, minimising the Within-Cluster Sum of Squares is equivalent to maximising:

$$\max_{\mathbf{U}} \operatorname{Tr}(\mathbf{X}^\top\mathbf{P}_\mathbf{U}\mathbf{X})$$

Geometrically, this corresponds to maximising the Between-Cluster Sum of Squares, i.e. the sum of squared distances between cluster centroids. Thus, minimising the internal variance of clusters is dual to maximising the separation between their centroids. While this combinatorial optimisation problem is NP-hard, local minima are efficiently found using the standard iterative algorithm proposed by Lloyd (1982) or MacQueen (1967).

2.3 Principal Component Analysis

In the context of dimensionality reduction, Principal Components Analysis (PCA) is a is a well-established procedure that seeks a low-dimensional projection of the data, retaining as much of the original variance as possible.

Formally we consider the column space of some loading matrix $\mathbf{A}$, where PCA aims to find such loadings which minimise the reconstruction error between the original data $\mathbf{X}$ and its projection:

$$\ell_{PCA}(\mathbf{A}) = \|\mathbf{X} - \mathbf{X}\mathbf{A}\mathbf{A}^\top\|_F^2 \tag{2.2}$$

Eckart and Young (1936) demonstrate that the optimal L-rank approximation of $\mathbf{X}$ (in terms of Frobenius distance to $\mathbf{X}$) is given by the the L leading eigenvectors of the covariance matrix $\mathbf{X}^\top\mathbf{X}$. These can be obtained via the truncated Singular Value Decomposition (SVD) of the data matrix.

Singular Value Decomposition (SVD) in this context serves as a computational means to derive a necessary set of vectors rather than a theoretical necessity. SVD is a foundational procedure which allows for any real data matrix $\mathbf{X}$ to be factorised as $\mathbf{X} = \mathbf{\Phi}\mathbf{\Sigma}\mathbf{\Psi}^\top$ where $\mathbf{\Phi}$ and $\mathbf{\Psi}$ are real orthogonal matrices and $\mathbf{\Sigma}$ is a diagonal matrix of singular values. Consider the following:

$$\mathbf{X}^\top\mathbf{X} = (\mathbf{\Phi}\mathbf{\Sigma}\mathbf{\Psi}^\top)^\top(\mathbf{\Phi}\mathbf{\Sigma}\mathbf{\Psi}^\top) = \mathbf{\Psi}\mathbf{\Sigma}^\top\mathbf{\Phi}^\top\mathbf{\Phi}\mathbf{\Sigma}\mathbf{\Psi}^\top = \mathbf{\Psi}\mathbf{\Sigma}^2\mathbf{\Psi}^\top$$

This expansion reveals that the columns of $\mathbf{\Psi}$ are exactly the eigenvectors of $\mathbf{X}^\top\mathbf{X}$. Therefore, SVD serves as a computationally stable alternative to extract the optimal loading matrix $\mathbf{A}$.

2.4 Flaws of Tandem Analysis

The term 'tandem analysis' was coined by Arabie and Hubert (1994) and the tandem approach involves first using PCA to find a projection $\mathbf{A}$ which minimises objective 2.2 and then apply the K-means algorithm on the low-dimensional projection $\mathbf{XA}$ to derive an optimal partition $\mathbf{U}$ in terms of objective 2.1 (where the objective considers $\mathbf{XA}$ rather than $\mathbf{X}$). Clearly, this separates dimensionality reduction and clustering into two separate processes.

While computationally convenient, this approach is theoretically flawed for clustering tasks. As established, PCA selects the subspace that maximises total data variance. However, the directions of maximum variance do not necessarily coincide with the directions of maximum cluster separation. This leads to a widely recognised vulnerability known as the cluster masking problem.

Masking variables are features that exhibit high variance but contain minimal information regarding the clustering structure. Because PCA seeks dimensions with high variance, it will preferentially retain these high-variance masking variables to form the reduced subspace. If instead the cluster structure is contained in the dimensions with low variance i.e. the 'noise' dimensions in the eyes of PCA, the tandem approach may discard the taxonomic information before the clustering step even begins.

To demonstrate this, we construct a synthetic dataset from a known low-dimensional clustering structure and consider the effect of dimensions containing high variance noise. A dataset of $N = 300$ observations across $P = 8$ total dimensions is generated, where the first two variables are formulated to contain the true clustering information. These two variables are sampled from one of three latent Gaussian clusters with centroids $(0, \sqrt{13}), (-3, -2)$ and $(3, -2)$ that are equidistant from the origin. Their covariances are distinctly specified as diag$(1.5, 0.2)$, diag$(0.2, 1.5)$ and diag$(0.5, 0.5)$ respectively so as to form three clusters of which one is warped along the first dimension, one warped along the second dimension as well as a spherical Gaussian cluster. The remaining six variables are composed using independent Gaussian noise with standard deviations $\sigma = 3$ and $\sigma = 1$ representing high and low variance masking variables which contain absolutely no information regarding cluster structure. This results in two 300×8 datasets, one containing relatively high-variance masking noise and the other containing relatively low-variance noise.

The tandem analysis procedure is carried out on these two datasets by computing the PCA transformations and applying the K-means clustering algorithm with $K = 3$ on the projection of the data onto the first two principal components, in order to obtain predicted cluster groupings. We plot the obtained results over the computed principal components below.

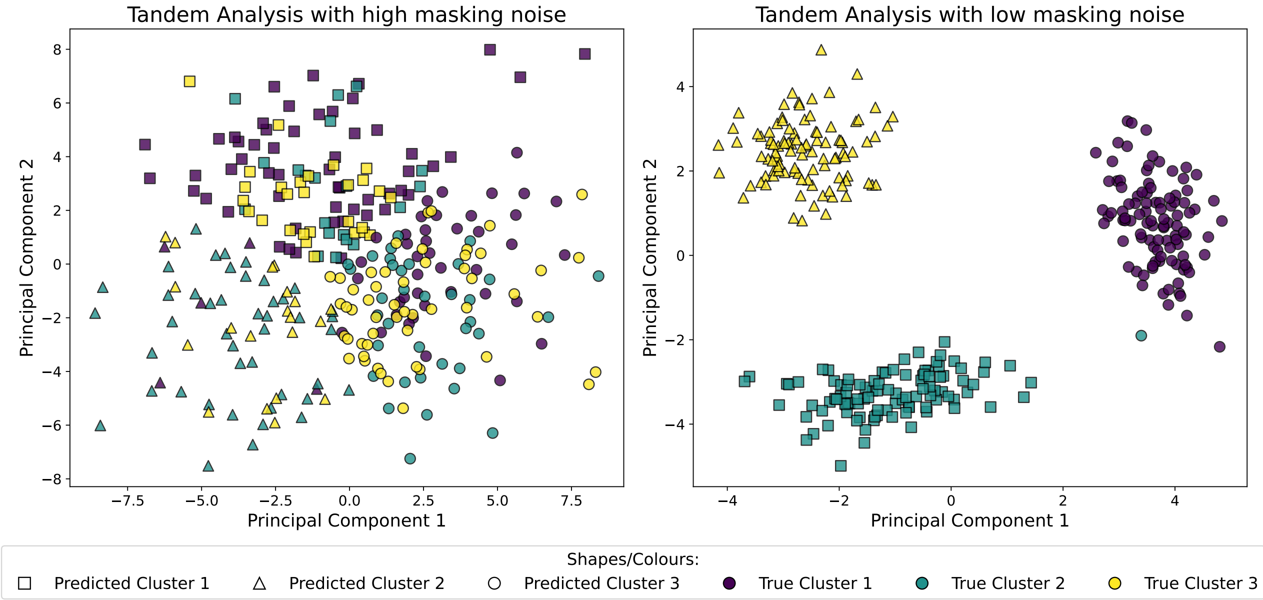

Figure 2.1 Tandem Analysis with high/low variance noise dimensions

As illustrated in Figure 2.1 (right), when the masking variables exhibit relatively low variance, this sequential methodology succeeds. Because the variance within the true taxonomic subspace is sufficiently large relative to the noise dimensions, PCA is able to correctly identify and retain the latent structural dimensions. This then allows the K-means algorithm to accurately recover the true cluster assignments.

In contrast, the presence of high-variance masking variables entirely distorts the results obtained through the tandem procedure, as demonstrated in Figure 2.1 (left). Because PCA seeks to maximise the variance retained, it constructs the principal components largely using the noise variables since in this case they dominate the variance within the dataset. This in turn forces the K-means algorithm to assign clusters within a subspace dominated by random noise dimensions, resulting in arbitrary and distorted cluster assignments. This outcome validates the cluster masking problem, showing that selecting a subspace strictly based on maximum variance provides no mathematical guarantee of retaining cluster separability.

In addition, we can inspect and analyse the underlying PCA loading matrices used to transform each dataset. Table 2.1 displays the coefficients assigned to each original variable to project onto the two-dimensional reduced subspace.

	High Variance		**Low Variance**	
Variable	PC1	PC2	PC1	PC2
Var 1	0.1168	-0.1182	**0.2080**	**0.9731**
Var 2	0.1253	-0.1695	**0.9751**	**-0.2132**
Var 3	**-0.3892**	-0.3150	0.0495	0.0152
Var 4	0.0552	**0.5232**	-0.0373	-0.0447
Var 5	-0.0056	**0.4790**	-0.0183	-0.0153
Var 6	**-0.5219**	-0.2026	0.0019	-0.0010
Var 7	-0.1136	**0.5148**	-0.0139	-0.0434
Var 8	**0.7286**	-0.2210	-0.0393	-0.0564

Table 2.1 PCA loading matrices for high and low variance masking scenarios

The loading values in Table 2.1 reveal the underlying reasons behind the vastly different plots in Figure 2.1. When the masking noise contains low variance, PCA assigns the majority of the weights to Variables 1 and 2, successfully isolating the true taxonomic signal while largely ignoring the irrelevant dimensions. However, when the masking variables contain high variance, PCA distributes its heaviest loadings across various masking variables. Moreover, the proportional weights of the clustering-relevant Variables 1 and 2 have vastly decreased, confirming that information from these Variables is discarded in the tandem analysis procedure in favour of the masking variables which leads to the distorted clustering space seen earlier.

Chapter 3

Simultaneous Dimensionality Reduction and Clustering Methods

3.1 Reduced K-means (RKM)

In an attempt to address the flaws of tandem analysis which can fail in the presence of high-variance noise, De Soete and Carroll (1994) introduce a procedure where the clustering and dimensionality reduction are coupled into a single objective function. Reduced K-means (RKM) seeks to minimise the reconstruction error of cluster centroids rather than maximise total variance. Specifically, they model the data matrix as:

$$\mathbf{X} = \mathbf{P}_{\mathbf{U}}\mathbf{X}\mathbf{A}\mathbf{A}^{\top} + \mathbf{E} \tag{3.1}$$

where $\mathbf{E}$ is an $N \times P$ error matrix. The term $\mathbf{X}\mathbf{A}\mathbf{A}^{\top}$ represents the reconstruction of the low-dimensional projection $\mathbf{X}\mathbf{A}$ back onto the original space, while $\mathbf{P}_{\mathbf{U}}$ projects these data points onto their respective cluster centroids. Therefore $\mathbf{E}$ contains information loss due to both dimensionality reduction and clustering.

The objective function is formulated as the squared Frobenius norm of these errors, minimised jointly over both the subspace projection matrix $\mathbf{A}$ and partition $\mathbf{U}$:

$$\ell_{RKM}(\mathbf{A}, \mathbf{U}) = \|\mathbf{X} - \mathbf{P}_{\mathbf{U}}\mathbf{X}\mathbf{A}\mathbf{A}^{\top}\|_F^2 \tag{3.2}$$

De Soete and Carroll (1994) also provide an alternating least-squares algorithm to find a local minimum for ℓ_{RKM}.
The alternating steps in Algorithm 1 highlight the fundamental differences between Reduced K-means and the tandem approach.

While the tandem approach fixes loadings $\mathbf{A}$ based on directions of total data variance, RKM dynamically updates the subspace using the SVD of the weighted cluster centroids (Lines 3–4). This strategy ensures that the subspaces are selected to maximise the between-cluster variance, prioritising the separation of the clusters over the overall spread of the data.

To see that this update of $\mathbf{A}$ does indeed reduce the objective, first note that $\mathbf{C}_w^{\top}\mathbf{C}_w = \mathbf{X}^{\top}\mathbf{P}_{\mathbf{U}}\mathbf{X}$ and therefore,

$$\begin{aligned}\ell_{RKM}(\mathbf{A}, \mathbf{U}) &= \mathrm{Tr}(\mathbf{X}^{\top}\mathbf{X}) - \mathrm{Tr}(\mathbf{A}^{\top}\mathbf{X}^{\top}\mathbf{P}_{\mathbf{U}}\mathbf{X}\mathbf{A}) \\ &= \mathrm{Tr}(\mathbf{X}^{\top}\mathbf{X}) - \mathrm{Tr}(\mathbf{A}^{\top}\mathbf{C}_w^{\top}\mathbf{C}_w\mathbf{A})\end{aligned}$$

Since $\mathrm{Tr}(\mathbf{X}^{\top}\mathbf{X})$ is constant, it is therefore sufficient to maximise the term $\mathrm{Tr}(\mathbf{A}^{\top}\mathbf{C}_w^{\top}\mathbf{C}_w\mathbf{A})$ over $\mathbf{A}$ in order to minimise the objective. This is achieved by taking the leading right singular vectors of $\mathbf{C}_w$ or alternatively, the leading eigenvectors of $\mathbf{C}_w^{\top}\mathbf{C}_w$.

Algorithm 1: Reduced K-means (RKM)

Input: $\mathbf{X} \in \mathbb{R}^{N \times P}$, the data matrix
K, the number of clusters
L, the target subspace dimension
ϵ, the convergence tolerance
`maxiter`, the maximum number of iterations

Output: Optimal partition $\mathbf{U}$ and projection matrix $\mathbf{A}$

Initialise partition $\mathbf{U}$ randomly;
`// Initialise objective value and iterations`
$\ell_{old} \leftarrow \infty$; `iter` $\leftarrow 1$;
while $|\ell_{old} - \ell_{new}| > \epsilon$ ***and*** `iter` $\leq$ `maxiter` **do**
  `// Fix U, Update A`
  `// Principal components of weighted cluster centroids`
  $\mathbf{C}_w \leftarrow (\mathbf{U}^\top \mathbf{U})^{-1/2} \mathbf{U}^\top \mathbf{X}$;
  $\mathbf{A} \leftarrow L$ leading eigenvectors of $\mathbf{C}_w^\top \mathbf{C}_w$;
  `// Fix A, Update U`
  $\mathbf{U} \leftarrow$ Apply K-means update step on $\mathbf{XA}$;
  `// Update objective values and iteration`
  $\ell_{old} \leftarrow \ell_{new}$;
  $\ell_{new} \leftarrow \ell_{RKM}(\mathbf{A}, \mathbf{U})$;
  `iter` $\leftarrow$ `iter` $+ 1$;
end
return $\mathbf{U}, \mathbf{A}$

The cluster membership matrix $\mathbf{U}$ is updated via K-means on the data projection on this optimised subspace. By coupling these steps, RKM forces the loadings and the cluster partition to iteratively refine each other, recovering cluster structures that may have otherwise been masked by the principal components of the raw data.

A significant practical limitation of this alternating optimisation strategy is the high sensitivity to initial conditions. It is important to note that we only have guarantee of converge to a local minimum rather than the global optimum, thus the final outputs $\mathbf{A}$ and $\mathbf{U}$ are dependent on the initial random states. However this can be mitigated by recognising that crucially, Algorithm 1 outlines only a single execution of this process. In practice, to avoid converging to a poor local minimum, the algorithm is run multiple times using different random initialisations and the outputs with the lowest objective value over these runs is chosen.

Revisiting the example explored in Section 2.4 regarding scenarios where tandem analysis can struggle to produce a desired or accurate result, we apply the Reduced K-means algorithm to the exact dataset containing high-variance masking variables.

As illustrated in Figure 3.1, RKM successfully recovers the three latent clusters, bypassing the cluster masking problem that caused issues within the tandem approach. We can further verify this by inspecting the weights of the computed loading matrix $\mathbf{A}$ in Table 3.1. Unlike standard PCA, which contained loadings dominated by the high variance noise variables, the simultaneous RKM procedure correctly assigns the majority of the weights to Variables 1 and 2, which contain the clustering information. This confirms that by jointly optimising the projection and clustering, RKM can in some cases accurately isolate variables relevant to an underlying cluster structure even in the presence of noisy masking variables.

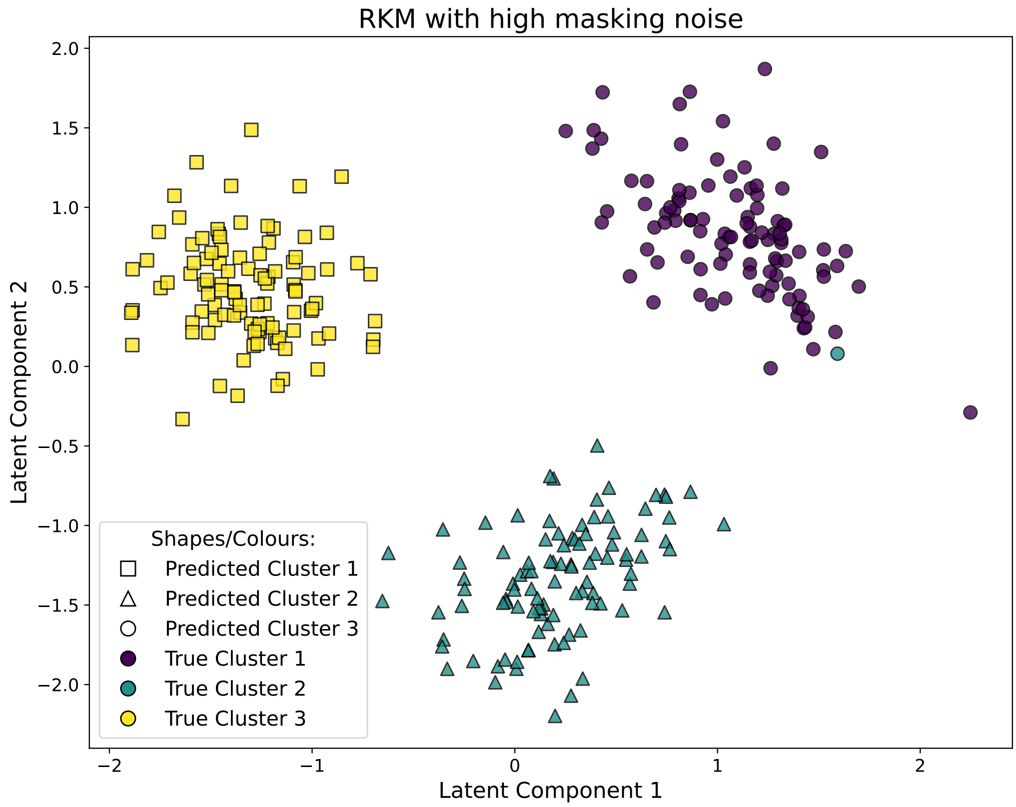

Figure 3.1 Reduced K-means with high variance noise dimensions

	Variable							
	Var 1	Var 2	Var 3	Var 4	Var 5	Var 6	Var 7	Var 8
Comp. 1	**-0.6111**	**-0.7763**	-0.0501	-0.0810	0.0284	0.0424	0.0265	-0.1074
Comp. 2	**0.7817**	**-0.6181**	-0.0448	-0.0014	-0.0584	-0.0354	0.0064	0.0141

Table 3.1 RKM loading matrix for the high variance masking scenario.

3.2 Factorial K-means (FKM)

While Reduced K-means addresses the decoupling issues of tandem analysis, Vichi and Kiers (2001) identify that it may still struggle to find underlying clustering structures in certain scenarios. These issues arise due to approach of De Soete and Carroll (1994), where the proposed model 3.1 considers the reconstruction error between objects and their low-dimensional cluster centroids. This approach can struggle when the original space contains much variance or noise in directions orthogonal to ones capturing the interesting clustering. Variance in such directions may contribute considerably to the distances between data points and the centroids, effectively penalising the algorithm if it decides to discard these directions and potentially leading it to select a subspace that captures this noise rather than the cluster structure.

To overcome this, Vichi and Kiers (2001) propose a method which they name Factorial K-means (FKM). They note that it seems more consistent to find a subspace where the projected data points have the smallest distances to centroids within the same subspace. By measuring the error with respect to the low-dimensional projection, FKM effectively ignores variance in orthogonal directions, ensuring that the subspace selection is driven exclusively by the clustering structure. The Factorial

K-means model assumes form:

$$\mathbf{X}\mathbf{A}\mathbf{A}^\top = \mathbf{P_U}\mathbf{X}\mathbf{A}\mathbf{A}^\top + \mathbf{E} \tag{3.3}$$

Similarly, the objective function it seeks to minimise takes form:

$$\ell_{FKM}(\mathbf{A}, \mathbf{U}) = \|\mathbf{X}\mathbf{A} - \mathbf{P_U}\mathbf{X}\mathbf{A}\|_F^2 \tag{3.4}$$

Note that it is clear by expanding that $\|\mathbf{X}\mathbf{A}\mathbf{A}^\top - \mathbf{P_U}\mathbf{X}\mathbf{A}\mathbf{A}^\top\|_F^2 = \|\mathbf{X}\mathbf{A} - \mathbf{P_U}\mathbf{X}\mathbf{A}\|_F^2$ due to the cyclic property of trace, which gives a simpler form for objective 3.4. Vichi and Kiers (2001) propose an alternating least-squares algorithm to solve this optimisation problem which we present in Algorithm 2.

Algorithm 2: Factorial K-means (FKM)

Input: $\mathbf{X} \in \mathbb{R}^{N \times P}$, the data matrix
K, the number of clusters
L, the target subspace dimension
ϵ, the convergence tolerance
`maxiter`, the maximum number of iterations

Output: Optimal partition $\mathbf{U}$ and projection matrix $\mathbf{A}$

Initialise partition $\mathbf{U}$ randomly;
`// Initialise objective value and iterations`
$\ell_{old} \leftarrow \infty$; `iter` $\leftarrow 1$;
while $|\ell_{old} - \ell_{new}| > \epsilon$ ***and*** `iter` $\leq$ `maxiter` **do**
    `// Fix U, Update A`
    $\mathbf{A} \leftarrow L$ leading eigenvectors of $\mathbf{X}^\top(\mathbf{P_U} - \mathbf{I}_N)\mathbf{X}$;
    `// Fix A, Update U`
    $\mathbf{U} \leftarrow$ Apply K-means update step on $\mathbf{X}\mathbf{A}$;
    `// Update objective values and iteration`
    $\ell_{old} \leftarrow \ell_{new}$;
    $\ell_{new} \leftarrow \ell_{FKM}(\mathbf{A}, \mathbf{U})$;
    `iter` $\leftarrow$ `iter` $+ 1$;
end
return $\mathbf{U}, \mathbf{A}$

In order to update $\mathbf{U}$ when $\mathbf{A}$ is fixed, the FKM objective 3.4 takes exactly the same form as the standard K-means objective 2.1 with $\mathbf{X}\mathbf{A}$ rather than $\mathbf{X}$. As such, it is clear why the K-means algorithm produces a desired and sufficient result.

The task of updating $\mathbf{A}$ is less straightforward and relies on the expansion

$$\begin{aligned}\|\mathbf{X}\mathbf{A} - \mathbf{P_U}\mathbf{X}\mathbf{A}\|_F^2 &= \|(\mathbf{I}_N - \mathbf{P_U})\mathbf{X}\mathbf{A}\|_F^2 \\ &= \mathrm{Tr}(\mathbf{A}^\top\mathbf{X}^\top(\mathbf{I}_N - \mathbf{P_U})\mathbf{X}\mathbf{A})\end{aligned}$$

Minimising $\mathrm{Tr}(\mathbf{A}^\top\mathbf{X}^\top(\mathbf{I}_N - \mathbf{P_U})\mathbf{X}\mathbf{A})$ is equivalent to maximising $\mathrm{Tr}(\mathbf{A}^\top\mathbf{X}^\top(\mathbf{P_U} - \mathbf{I}_N)\mathbf{X}\mathbf{A})$ and recalling that $\mathbf{A}$ is orthonormal, we can use a well-known matrix optimisation result (see Ten Berge (1993)). The minimum is attained when the columns of $\mathbf{A}$ are the leading eigenvectors of the matrix $\mathbf{X}^\top(\mathbf{I}_N - \mathbf{P_U})\mathbf{X}$.

These updates of $\mathbf{U}$ and $\mathbf{A}$ monotonically decrease the bounded loss function 3.4, therefore FKM converges to at least a local optimum. However due to the binary constraint on $\mathbf{U}$, this method can once again be sensitive to local optima based on the initial values of $\mathbf{U}$. As such, Vichi and Kiers (2001) recommend to either use many randomly started runs and retain only the solution with the lowest objective value or to use a good rational start, for instance via K-means or analysis of the data.

These methods rely on the pre-selection of number of clusters K and clustering subspace dimensions L. De Soete and Carroll (1994) note that the K cluster centroids always define a $(K-1)$-dimensional subspace and suggest that L should be set smaller than $\min(K-1, M)$ to achieve dimensionality reduction. Vichi and Kiers (2001) also suggest first choosing the number of clusters K and then choosing dimensionality as $K-1$.

3.3 Generalised Reduced Clustering (GRC)

Both Reduced K-means and Factorial K-means introduce the idea of a single optimisation criterion for conducting dimension reduction and clustering at the same time, and provide easy-to-implement algorithms which have been found to perform better than the tandem approach. Timmerman et al. (2010) provide theoretical and empirical comparisons of the two methods which are generally useful for simultaneous dimensionality reduction and clustering in many situations.

However Yamamoto and Hwang (2014) note a drawback in these two methods which originates from the way they approach dimensionality reduction. These two methods are likely to fail if there exists noise variables irrelevant to the clustering structure that are highly correlated between themselves. Both RKM and FKM assume that the information discarded during dimensionality reduction consists purely of unstructured noise. However, in practice it is not uncommon to observe that many irrelevant variables are correlated to some extent, and variance-based methods like RKM and FKM may mistake this irrelevant variation as part of the optimal subspace. For example, consider a dataset where the true objective is to cluster countries into distinct climate zones based on weather-based metrics. If the dataset concurrently includes a small number of economic indicators such as GDP, national debt, inflation rates etc. then this block of economic variables will be highly correlated between each other while failing to contribute information relevant to the desired clustering.

To directly address this issue, Yamamoto and Hwang (2014) propose a general framework that combines simultaneous reduction and clustering methods (as in RKM/FKM) with subspace separation techniques. This method, called Generalised Reduced Clustering (GRC), extends previous notions to explicitly distinguish between relevant and irrelevant feature sets of the data.

The model further separates the reduced subspace of dimension L into two smaller subspaces of dimension L_C and L_D with $L_C + L_D = L$. Overall the original P-dimensional space is split into three. In particular, we assume a relevant L_C-subspace containing the cluster structure and an irrelevant but structured L_D-subspace containing correlated noise. The remaining $P-L$-subspace is assumed to contain random noise.

Moreover, the loading matrix $\mathbf{A}$ is divided into two parts: $\mathbf{A} = \left(\mathbf{A}_C | \mathbf{A}_D\right)$, where $\mathbf{A}_C$ and $\mathbf{A}_D$ are $P \times L_C$ and $P \times L_D$ loading matrices that each span subspace of relevant and disturbing variables respectively.

This leads to the formulation of the GRC objective function:

$$\ell_{GRC}(\mathbf{A}, \mathbf{U};\, L_D, \rho_1, \rho_2) = \|\mathbf{X} - \mathbf{X}\mathbf{A}\mathbf{A}^\top\|_F^2 + \rho_1\|\mathbf{X}\mathbf{A}_C - \mathbf{P}_\mathbf{U}\mathbf{X}\mathbf{A}_C\|_F^2 + \rho_2\|\mathbf{P}_\mathbf{U}\mathbf{X}\mathbf{A}_C\|_F^2 \tag{3.5}$$

where $\rho_1, \rho_2 \in \mathbb{R}$ are user-specified parameters with $\rho_1 > \rho_2$. This objective consists of three components. The first component $\|\mathbf{X} - \mathbf{X}\mathbf{A}\mathbf{A}^\top\|_F^2$ corresponds to loss incurred by dimensionality reduction and is independent of the clustering step. The term $\|\mathbf{X}\mathbf{A}_C - \mathbf{P}_\mathbf{U}\mathbf{X}\mathbf{A}_C\|_F^2$ is of a similar form to the FKM objective 3.4, except that the data is projected onto the reduced space consisting of relevant variables $\mathbf{A}_C$ rather than the entire reduced space $\mathbf{A}$. The third component $\|\mathbf{P}_\mathbf{U}\mathbf{X}\mathbf{A}_C\|_F^2$ is the between-cluster variance in the relevant subspace. By weighting these components via ρ_1 and ρ_2, the GRC objective effectively balances the trade-off between minimising the reconstruction error (as in PCA) while minimising within-cluster spread and maximising cluster separation.

Note that when $L_D = 0$, we have $\mathbf{A} \equiv \mathbf{A}_C$ so the problem is similar to RKM/FKM where there is no notion of subspace separation and the whole reduced subspace is considered relevant. Particularly we note that minimisation of ℓ_{GRC} with parameters $\ell_{GRC}(\mathbf{A}, \mathbf{U};\ 0, 1, 0)$ reduces to $\ell_{RKM}(\mathbf{A}, \mathbf{U})$ and $\ell_{GRC}(\mathbf{A}, \mathbf{U};\ 0, 2, 1)$ is equivalent to $\ell_{FKM}(\mathbf{A}, \mathbf{U})$.

These results can easily be derived, consider the RKM case:

$$\begin{aligned}
\ell_{GRC}(\mathbf{A}, \mathbf{U};\ 0, 1, 0) &= \|\mathbf{X} - \mathbf{X}\mathbf{A}\mathbf{A}^\top\|_F^2 + \|\mathbf{X}\mathbf{A} - \mathbf{P}_\mathbf{U}\mathbf{X}\mathbf{A}\|_F^2 \\
&= \mathrm{Tr}\left((\mathbf{X} - \mathbf{X}\mathbf{A}\mathbf{A}^\top)^\top(\mathbf{X} - \mathbf{X}\mathbf{A}\mathbf{A}^\top)\right) + \mathrm{Tr}\left((\mathbf{X}\mathbf{A} - \mathbf{P}_\mathbf{U}\mathbf{X}\mathbf{A})^\top(\mathbf{X}\mathbf{A} - \mathbf{P}_\mathbf{U}\mathbf{X}\mathbf{A})\right) \\
&= \left[\mathrm{Tr}(\mathbf{X}^\top\mathbf{X}) - \mathrm{Tr}(\mathbf{X}^\top\mathbf{X}\mathbf{A}\mathbf{A}^\top)\right] + \left[\mathrm{Tr}(\mathbf{X}^\top\mathbf{X}\mathbf{A}\mathbf{A}^\top) - \mathrm{Tr}(\mathbf{X}^\top\mathbf{P}_\mathbf{U}\mathbf{X}\mathbf{A}\mathbf{A}^\top)\right] \\
&= \mathrm{Tr}(\mathbf{X}^\top\mathbf{X}) - 2\,\mathrm{Tr}(\mathbf{X}^\top\mathbf{P}_\mathbf{U}\mathbf{X}\mathbf{A}\mathbf{A}^\top) + \mathrm{Tr}(\mathbf{A}\mathbf{A}^\top\mathbf{X}^\top\mathbf{P}_\mathbf{U}{}^\top\mathbf{P}_\mathbf{U}\mathbf{X}\mathbf{A}\mathbf{A}^\top) \\
&= \mathrm{Tr}\left((\mathbf{X} - \mathbf{P}_\mathbf{U}\mathbf{X}\mathbf{A}\mathbf{A}^\top)^\top(\mathbf{X} - \mathbf{P}_\mathbf{U}\mathbf{X}\mathbf{A}\mathbf{A}^\top)\right) \\
&= \|\mathbf{X} - \mathbf{P}_\mathbf{U}\mathbf{X}\mathbf{A}\mathbf{A}^\top\|_F^2 \\
&= \ell_{RKM}(\mathbf{A}, \mathbf{U})
\end{aligned}$$

This also shows that,

$$\ell_{RKM}(\mathbf{A}, \mathbf{U}) = \|\mathbf{X} - \mathbf{X}\mathbf{A}\mathbf{A}^\top\|_F^2 + \ell_{FKM}(\mathbf{A}, \mathbf{U})$$

This decomposition captures the mathematical relationship between the two methods, where the RKM objective is simply the FKM clustering loss with an additional reconstruction error that is completely independent of the cluster partition. This highlights why RKM forces a compromise between preserving overall data variance and separating clusters, whereas FKM strictly optimises for cluster separation within the subspace. Similarly for the FKM parameters, since minimisation is independent of the constant $\|\mathbf{X}\|_F^2$:

$$\begin{aligned}
\ell_{GRC}(\mathbf{A}, \mathbf{U};\ 0, 2, 1) &= \|\mathbf{X} - \mathbf{X}\mathbf{A}\mathbf{A}^\top\|_F^2 + 2\|\mathbf{X}\mathbf{A} - \mathbf{P}_\mathbf{U}\mathbf{X}\mathbf{A}\|_F^2 + \|\mathbf{P}_\mathbf{U}\mathbf{X}\mathbf{A}\|_F^2 \\
&= \left[\mathrm{Tr}(\mathbf{X}^\top\mathbf{X}) - \mathrm{Tr}(\mathbf{X}^\top\mathbf{X}\mathbf{A}\mathbf{A}^\top)\right] + 2\left[\mathrm{Tr}(\mathbf{A}^\top\mathbf{X}^\top\mathbf{X}\mathbf{A}) - \mathrm{Tr}(\mathbf{A}^\top\mathbf{X}^\top\mathbf{P}_\mathbf{U}\mathbf{X}\mathbf{A})\right] \\
&\qquad + \mathrm{Tr}(\mathbf{A}^\top\mathbf{X}^\top\mathbf{P}_\mathbf{U}\mathbf{X}\mathbf{A}) \\
&= \mathrm{Tr}(\mathbf{X}^\top\mathbf{X}) - \mathrm{Tr}(\mathbf{X}^\top\mathbf{X}\mathbf{A}\mathbf{A}^\top) + 2\,\mathrm{Tr}(\mathbf{A}^\top\mathbf{X}^\top\mathbf{X}\mathbf{A}) \\
&\qquad - 2\,\mathrm{Tr}(\mathbf{A}^\top\mathbf{X}^\top\mathbf{P}_\mathbf{U}\mathbf{X}\mathbf{A}) + \mathrm{Tr}(\mathbf{A}^\top\mathbf{X}^\top\mathbf{P}_\mathbf{U}\mathbf{X}\mathbf{A}) \\
&= \mathrm{Tr}(\mathbf{X}^\top\mathbf{X}) + \left[\mathrm{Tr}(\mathbf{X}^\top\mathbf{X}\mathbf{A}\mathbf{A}^\top) - \mathrm{Tr}(\mathbf{X}^\top\mathbf{P}_\mathbf{U}\mathbf{X}\mathbf{A}\mathbf{A}^\top)\right] \\
&= \|\mathbf{X}\|_F^2 + \|\mathbf{X}\mathbf{A} - \mathbf{P}_\mathbf{U}\mathbf{X}\mathbf{A}\|_F^2 \\
&= \|\mathbf{X}\|_F^2 + \ell_{FKM}(\mathbf{A}, \mathbf{U})
\end{aligned}$$

Yamamoto and Hwang (2014) also provide an alternating least-squares algorithm to minimise 3.5 over $\mathbf{A}, \mathbf{U}$ given parameter values (L_D, ρ_1, ρ_2) for one run/random start, as seen in Algorithm 3.

The task of updating $\mathbf{U}$ is similar to the FKM and RKM case, supported by a lemma from Yamamoto and Hwang (2014) which states that for $\rho_1 > \rho_2$, the minimisation of $\ell_{GRC}(\mathbf{A}, \mathbf{U})$ over $\mathbf{U}$ is equivalent to the minimisation of $\ell_{FKM}(\mathbf{A}_C, \mathbf{U})$ over $\mathbf{U}$.
Updating $\mathbf{A}$ (Lines 3-7) relies on the gradient projection algorithm with an orthogonal constraint (see Jennrich (2001)). The variables $\mathbf{G}_C$, $\mathbf{G}_D$ in the algorithm correspond to $\partial\ell_{GRC}/\partial\mathbf{A}_C$, $\partial\ell_{GRC}/\partial\mathbf{A}_D$ respectively.

The Generalised Reduced Clustering framework introduces additional parametric complexity via the noise subspace dimension L_D and the penalty weights ρ_1, ρ_2. To address this, Yamamoto and Hwang (2014) suggest using a cross-validation procedure provided by Wang (2010), particularly *'cross-validation with averaging'*. Wang (2010) also discusses challenges in estimating the number of clusters and evaluates existing approaches to the task of assessing clustering quality, which in itself is a widely recognised problem due to the lack of an objective measure for such comparisons. The selection of L_C however, coincides with the selection of L in RKM/FKM and similar ideas can be used for this task.

Algorithm 3: Generalised Reduced Clustering (GRC)

Input: $\mathbf{X} \in \mathbb{R}^{N \times P}$, the data matrix
K, the number of clusters
L, the target subspace dimension
L_D, the distorting variables subspace dimension
(ρ_1, ρ_2), parameters controlling GRC objective
α, step size for gradient projection
ϵ, the convergence tolerance
`maxiter`, the maximum number of iterations

Output: Optimal partition $\mathbf{U}$ and projection matrix $\mathbf{A}$

Initialise partition $\mathbf{U}$ randomly;
`// Initialise objective value and iterations`
$\ell_{old} \leftarrow \infty$; `iter` $\leftarrow 1$;
while $|\ell_{old} - \ell_{new}| > \epsilon$ ***and*** `iter` $\leq$ `maxiter` **do**
  `// Fix U, Update A`
  `// Gradient Projection algorithm on A =(Ac | Ad)`
  $\mathbf{G}_C \leftarrow -2(1-\rho_1)\mathbf{X}^\top \mathbf{X}\mathbf{A}_C - 2(\rho_1 - \rho_2)\mathbf{X}^\top \mathbf{P}_\mathbf{U}\mathbf{X}\mathbf{A}_C$;
  $\mathbf{G}_D \leftarrow -2\mathbf{X}^\top \mathbf{X}\mathbf{A}_D$;
  $\mathbf{G} \leftarrow (\mathbf{G}_C | \mathbf{G}_D)$
  `// Move A and project to orthogonal space`
  $\mathbf{U\Sigma V}^\top \leftarrow \text{svd}(\mathbf{A} - \alpha\mathbf{G})$;
  $\mathbf{A} \leftarrow \mathbf{UV}^\top$;
  `// Fix A, Update U`
  $\mathbf{U} \leftarrow$ Apply K-means update step on $\mathbf{X}\mathbf{A}_C$
  `// Update objective values and iteration`
  $\ell_{old} \leftarrow \ell_{new}$;
  $\ell_{new} \leftarrow \ell_{GRC}(\mathbf{A}, \mathbf{U};\, L_D, \rho_1, \rho_2)$;
  `iter` $\leftarrow$ `iter` $+ 1$;
end
return $\mathbf{U}, \mathbf{A}$

Chapter 4

Disjointness Considerations

The Reduced K-means and Factorial K-means procedures are capable of reliably identifying low-dimensional subspaces which reflect an underlying cluster structure. While this is incredibly beneficial from a mathematical perspective, Vichi (2002) and Vichi and Saporta (2009) recognised that the resulting loading matrix $\mathbf{A}$ and its columns often lacks interpretability as some observable features contribute to multiple distinct components in the reduced space. This means that while the dimensions characterised by $\mathbf{A}$ are mathematically relevant to an underlying cluster structure, it remains difficult to gauge the effect of observed variables, and their relationships and contribution to the clustering.

For instance, consider the objective of clustering cinematic releases into distinct success profiles using high-dimensional data. If a dataset contains both financial metrics (e.g., production budget, marketing spend, box office gross) and critical reception metrics (e.g., average review scores, audience ratings, total award nominations) then these two groups of metrics naturally split into different categories. When methods such as RKM and FKM are employed to construct a reduced subspace, their reliance on continuous eigenvalue decompositions leaves them susceptible to rotational indeterminacy. Essentially, these algorithms can converge upon an arbitrary rotation of the subspace where the weights of the corresponding loading matrix $\mathbf{A}$ are densely dispersed across all variables for each latent component. Therefore, while these configurations may yield informative clustering partitions, an analyst cannot definitively state whether a specific latent dimension relates to a film's financial profitability or its critical acclaim, thus losing a level of interpretability.

To address this, Vichi (2002) introduce a disjointness constraint on the loading matrix $\mathbf{A}$ in the FKM model 3.2 and Vichi and Saporta (2009) later extended this constraint onto the RKM model 3.1 and also introduced an efficient algorithm to solve this problem.

The additional constraints on the $P \times L$ orthonormal matrix $\mathbf{A}$ are:

$$\sum_{p=1}^{P} (a_{pq} a_{pr})^2 = 0 \quad \text{for} \quad 1 \leq q, r \leq L,\, q \neq r \tag{4.1}$$

This constraint enforces the requirement that across any row of $\mathbf{A}$, at most one of the entries are non-zero. Therefore each variable can contribute to at most one of the components that characterise the reduced cluster subspace. From here on, we will refer to these constraints as the disjointness constraints. To visualise this, consider the following example disjoint loading matrix $\mathbf{A}$:

$$\mathbf{A} = \begin{pmatrix} \sqrt{1/2} & 0 \\ 0 & \sqrt{2/3} \\ 0 & \sqrt{1/3} \\ \sqrt{1/2} & 0 \end{pmatrix}$$

This represents a projection from a four-dimensional space to a two-dimensional space spanned by $\{(\sqrt{1/2} \quad 0 \quad 0 \quad \sqrt{1/2}), (0 \quad \sqrt{2/3} \quad \sqrt{1/3} \quad 0)\}$, the two columns. In particular we note that variables one and four contribute only to the first component and variables two and three only contribute to the second component. This strict separation allows for the interpretability of the reduced space, as we can directly attribute findings in the low-dimensional projection to specific variables from the original high-dimensional data.

4.1 Clustering and Disjoint Principal Component Analysis (CDPCA)

First introduced by Vichi and Saporta (2009), the CDPCA procedure applies the disjointness constraint on the Reduced K-means (RKM) objective. That is, the CDPCA obejctive takes the form:

$$\ell_{CDPCA}(\mathbf{A}, \mathbf{U}) = \ell_{RKM}(\mathbf{A}, \mathbf{U}) = \|\mathbf{X} - \mathbf{P}_{\mathbf{U}}\mathbf{X}\mathbf{A}\mathbf{A}^\top\|_F^2 \quad \text{subject to constraints 4.1} \tag{4.2}$$

In essence, the algorithm seeks to identify both the optimal clustering of observations and the disjoint grouping of variables that minimise the RKM objective. To operationalise this set of constraints, we introduce a binary $P \times L$ component membership matrix $\mathbf{V}$, defined such that $v_{pl} = 1$ if variable p loads onto latent component l, and $v_{pl} = 0$ otherwise. Therefore, for a fixed variable $p \in \{1, \dots, P\}$, we require that $\sum_{l=1}^{L} v_{pl} = 1$ in order to satisfy the disjointness constrains.

Note that $\mathbf{V}$ is introduced for computational ease rather than mathematical necessity. The constrained objective 4.2 is mathematically self-sufficient however introducing $\mathbf{V}$ allows us to explicitly enforce and optimise against the disjointness constraints via combinatorial methods. This will become clear as a modified version of Vichi and Saporta (2009)'s algorithm is presented in Algorithm 4.

The core ideas behind the $\mathbf{A}$ and $\mathbf{U}$ updates remain the same as those in Algorithm 1 for RKM, with the $\mathbf{U}$ update taking the exact same form. However, the columns of $\mathbf{A}$ are updated sequentially based on a fixed variable-to-component partition rather than updating it as a whole. In addition, this implementation of the $\mathbf{V}$ update (lines 12-15) utilises a greedy heuristic that differs from the original procedure outlined by Vichi and Saporta (2009).

In the original formulation, Vichi and Saporta (2009) rely on a similar sequential block descent procedure along the rows of $\mathbf{V}$ (i.e. each variable p) but outline a much more mathematically stricter update at each row. For a fixed variable p, their approach requires temporary assignments across all L components. Each temporary assignment $p \to l$ induces a change in $\mathbf{A}$ via an eigenvector computation and this modified $\mathbf{A}$ is used to re-evaluate the global CDPCA objective. Then variable p is assigned to the component l which yielded the lowest objective value across $1, \dots, L$. According to the authors, their stronger algorithm monotonically decreases the objective function and will converge to a stationary point which can be expected to be at least a local minimum.

However, this nested approach comes at an incredibly exhaustive computational cost due to the $\mathcal{O}(PL)$ eigenvector computations needed at each iteration. To reduce computational complexity, Algorithm 4 decouples the combinatorial assignment from the continuous subspace update. The loadings matrix $\mathbf{A}$ is assumed to be locally constant during the variable partition phase and therefore the use of temporary reassignments and eigenvector computations are removed. Instead the absolute values of a target matrix $\mathbf{C}_w^\top \mathbf{C}_w \mathbf{A}$ are used, where its elements quantify the scalar projection of variables onto latent factors. By greedily assigning each variable to the factor that maximises this projection, the algorithm makes locally optimal choices in order to decrease the objective value. Overall, while the relaxation of the original constraints results in the loss of strict monotonic convergence guarantees, the number of eigenvector computations comes down massively to $\mathcal{O}(L)$ and in practice the algorithm still drives the objective in a direction towards a minimum. Further theoretical motivation for thi approach stems from other established ALS frameworks (see Kiers (2002); de Leeuw (1994)) where complex joint-minimisation problems are split up by holding one parameter set fixed while updating another.

Algorithm 4: Clustering and Disjoint Principal Component Analysis (CDPCA)

Input: $\mathbf{X} \in \mathbb{R}^{N \times P}$, the data matrix
K, the number of clusters
L, the target subspace dimension
ϵ, the convergence tolerance
`maxiter`, the maximum number of iterations
Output: Optimal partition $\mathbf{U}$, projection matrix $\mathbf{A}$

Initialise cluster partition $\mathbf{U}$ randomly;
Initialise variable partition $\mathbf{V}$ randomly;
`// Initialise objective value and iterations`
$\ell_{old} \leftarrow \infty$; `iter` $\leftarrow 1$;
while $|\ell_{old} - \ell_{new}| > \epsilon$ ***and*** `iter` $\leq$ `maxiter` **do**
    `// Fix A, Update U`
    $\mathbf{U} \leftarrow$ Apply K-means update step on $\mathbf{XA}$;
    `// Pre-compute weighted cluster centroids`
    $\mathbf{C}_w \leftarrow (\mathbf{U}^\top \mathbf{U})^{-1/2} \mathbf{U}^\top \mathbf{X}$;
    `// Fix U and V, Update A`
    for *component* $l = 1$ **to** L **do**
        `// Indices of variables loading onto factor` l
        $\mathcal{I}_l \leftarrow \{p \mid v_{pl} = 1\}$;
        $\mathbf{C}_{sub} \leftarrow$ Columns of $\mathbf{C}_w$ corresponding to $\mathcal{I}_l$;
        `// Assign` l^{th} `column of` $\mathbf{A}$
        $\mathbf{A}_{\mathcal{I}_l, l} \leftarrow$ Leading eigenvector of $\mathbf{C}_{sub}^\top \mathbf{C}_{sub}$;
        $\mathbf{A}_{\setminus \mathcal{I}_l, l} \leftarrow 0$;
    end
    `// Fix U and A, Update V`
    for *variable* $p = 1$ **to** P **do**
        `// Largest absolute value over row` p
        $l^* \leftarrow \text{argmax}_{1 \leq l \leq L} |\mathbf{C}_w^\top \mathbf{C}_w \mathbf{A}|_{p,l}$;
        $v_{pl^*} \leftarrow 1$;
        $v_{pm} \leftarrow 0 \quad \text{for } m \neq l^*$;
    end
    `// Update objective values and iteration`
    $\ell_{old} \leftarrow \ell_{new}$;
    $\ell_{new} \leftarrow \ell_{CDPCA}(\mathbf{A}, \mathbf{U})$;
    `iter` $\leftarrow$ `iter` $+ 1$;
end
return $\mathbf{U}, \mathbf{A}$

An alternative consideration which incorporates a mixture of the two ideas could be to use a random search to sample without replacement $\tilde{p} \in \{1, \ldots, P\}$ and update $\mathbf{A}$ after each subsequent row-update of $\mathbf{V}$. This results in $\mathcal{O}(L + P)$ eigenvector computations (of $\mathbf{A}$) at each iteration. This however is not implemented in this case.

Having established the theoretical framework of CDPCA, we aim to empirically evaluate its ability to improve upon the baseline set by RKM in the presence of a variables that are naturally paritioned. To achieve this, we construct a synthetic dataset by assuming an underlying latent cluster structure and projecting this space this back onto a high-dimensional space via the transpose of a disjoint loading matrix $\mathbf{A}$. Given a dataset of high-dimensional observations, we aim to compare the abilities of RKM and CDPCA in both retrieving accurate cluster groupings and recovering the true loading matrix. Since the issues relating to masking variables have been explored in Section 2.4, we consider a scenario without masking noise. An $L = 2$ latent space is constructed containing $K = 4$ clusters with centroids located at the vertices of a square about the origin, with $N = 300$ observations projected onto a $P = 8$ dimensional space. For simplicity, the true loading matrix splits the variables evenly across the two components with equal weights, as seen in Table 4.1.

The recovered cluster grouping are plotted below in Figure 4.1. As illustrated, both the RKM and CDPCA procedures are highly competent at recovering a low-dimensional subspace that accurately captures the latent cluster structure. Visually, the partitions are very similar, and both methods are competent at isolating the four groups. However, the fundamental distinction between the two frameworks become apparent when inspecting their corresponding recovered loading matrices, presented in Table 4.1. RKM disperses its loading weights across all available variables, creating a overlapping configuration that loses interpretability in terms of the original variables. In contrast, the basis vectors derived by CDPCA align perfectly with the true variable-component partitions and CDPCA accurately recovers the loading weights of the true projection matrix. This shows that although both models are able to obtain accurate clusters, the specific loadings extracted by CDPCA provide a subspace that is much more interpretable as it avoids spreading loadings out across variables.

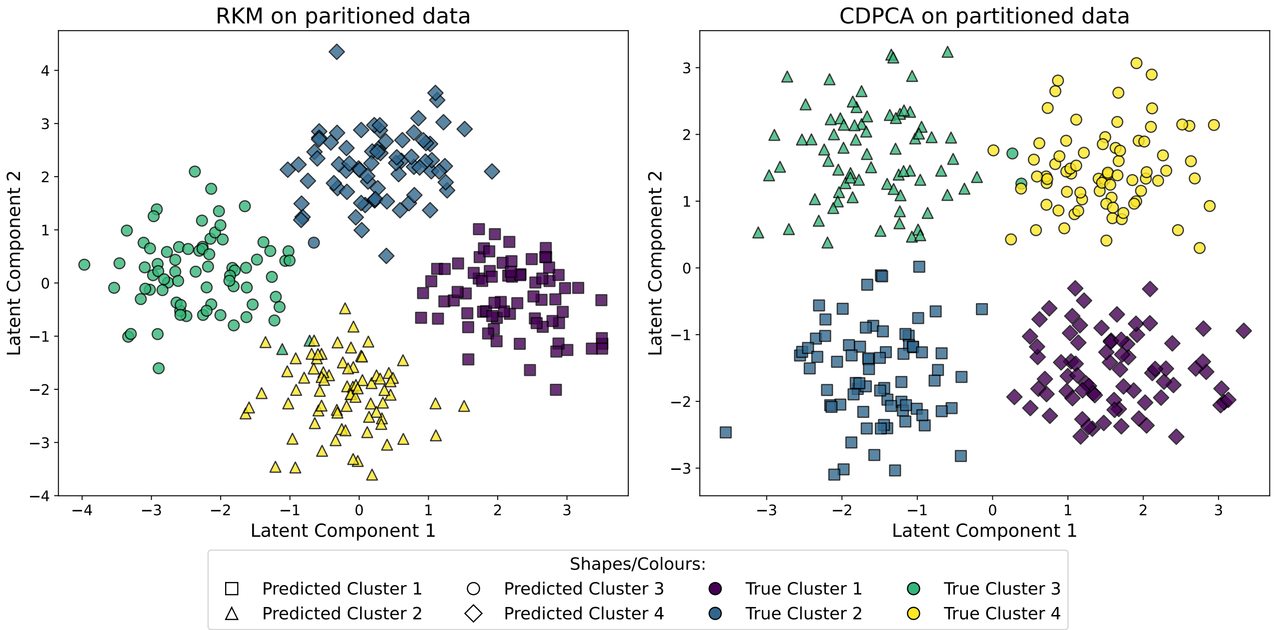

Figure 4.1 RKM and CDPCA projections of synthetic observations, generated with a disjoint loading matrix

	True		RKM		CDPCA	
Variable	Comp. 1	Comp. 2	Comp. 1	Comp. 2	Comp. 1	Comp. 2
Var 1	0.5000	0	0.2601	-0.4254	0	-0.4969
Var 2	0.5000	0	0.3029	-0.4080	0	-0.5086
Var 3	0.5000	0	0.2724	-0.4217	0	-0.5013
Var 4	0.5000	0	0.2957	-0.3940	0	-0.4931
Var 5	0	0.5000	-0.4182	-0.2894	-0.5089	0
Var 6	0	0.5000	-0.4079	-0.2734	-0.4916	0
Var 7	0	0.5000	-0.4124	-0.2767	-0.4972	0
Var 8	0	0.5000	-0.4094	-0.2907	-0.5021	0

Table 4.1 Comparison of the true loading matrix against the matrices recovered by RKM and CDPCA

4.2 Disjoint Factorial K-means (DFKM)

CDPCA is a clear example of how simultaneous dimensionality reduction and clustering methods can be modified to identify highly-interpretable subspaces with simpler, sparse and disjoint dimensions. However, because CDPCA operates on optimising a constrained RKM objective, it is only natural that it inherits the fundamental theoretical vulnerabilities of the RKM procedure. Namely, if a dataset contains high-variance dimensions that are orthogonal to the true clustering structure, the CDPCA model may struggle. This could result in retaining noisy and irrelevant variables in the reduced subspace as mentioned in Section 3.2. As such, this raises the question of whether the disjointness constraints 4.1 can be applied to the FKM model in order to overcome some of these shortcomings.

The work by Vichi (2002), long before the development of CDPCA, begins to detail some of the theoretical aspects behind such a model but they do not provide a similar ALS procedure. Instead they propose using sequential quadratic programming methods. We attempt to bridge this gap and introduce a framework that incorporates disjointness into the FKM model while also providing a simpler algorithm to tackle such problems. As we attempt this, there are some dangerous pitfalls within the original FKM framework that become much more apparent with the introduction of harsher objective constraints.

To begin with, recall the FKM objective which forms the base of the proposed *Disjoint Factorial K-means* objective:

$$\ell_{RKM}(\mathbf{A}, \mathbf{U}) = \ell_{FKM}(\mathbf{A}, \mathbf{U}) = \|\mathbf{XA} - \mathbf{P_U XA}\|_F^2 \quad \text{subject to constraints 4.1} \tag{4.3}$$

4.3 Sensitivity to Low Subspace Variance

As previously established, the RKM objective can be rewritten as the total variance of the dataset minus the explained between-cluster variance:

$$\ell_{RKM}(\mathbf{A}, \mathbf{U}) = \mathrm{Tr}(\mathbf{X}^\top \mathbf{X}) - \mathrm{Tr}(\mathbf{A}^\top \mathbf{X}^\top \mathbf{P_U XA})$$

Noticing that the first term, $\mathrm{Tr}(\mathbf{X}^\top \mathbf{X})$, acts as a fixed global constant independent of the loading matrix $\mathbf{A}$, it is deemed sufficient and equivalent to maximise $\mathrm{Tr}(\mathbf{A}^\top \mathbf{X}^\top \mathbf{P_U XA})$ in order to minimise the objective. When $\mathbf{U}$ is held constant, this problem leads to stable eigenvector solutions via SVD and therefore provides a highly stable theoretical framework for CDPCA to enforce discrete constraints by splitting into sub-blocks.

On the other hand, we can try to obtain a similar form for the FKM objective:

$$\begin{aligned}\ell_{FKM}(\mathbf{A},\mathbf{U}) = \|\mathbf{X}\mathbf{A}-\mathbf{P}_{\mathbf{U}}\mathbf{X}\mathbf{A}\|_F^2 &= \|(\mathbf{I}_N-\mathbf{P}_{\mathbf{U}})\mathbf{X}\mathbf{A}\|_F^2 \\ &= \mathrm{Tr}(\mathbf{A}^\top\mathbf{X}^\top(\mathbf{I}_N-\mathbf{P}_{\mathbf{U}})\mathbf{X}\mathbf{A}) \\ &= \mathrm{Tr}(\mathbf{A}^\top\mathbf{X}^\top\mathbf{X}\mathbf{A}) - \mathrm{Tr}(\mathbf{A}^\top\mathbf{X}^\top\mathbf{P}_{\mathbf{U}}\mathbf{X}\mathbf{A})\end{aligned}$$

A crucial observation is that within the FKM expansion, the total variance term has been replaced by $\mathrm{Tr}(\mathbf{A}^\top\mathbf{X}^\top\mathbf{X}\mathbf{A})$ which is now dependent on the projection $\mathbf{X}\mathbf{A}$. This significantly alters the optimisation space as the algorithm can achieve a mathematically 'optimal' objective value by selecting a subspace where the total projected variance is exceptionally small.

In other words, $\mathrm{Tr}(\mathbf{A}^\top\mathbf{X}^\top\mathbf{X}\mathbf{A})$ and in turn the objective can be made significantly smaller by choosing a subspace (represented by the span of $\mathbf{A}$) with low-variance dimensions, without regard for separation between clusters. By artificially shrinking the entire projected space, the objective function can be trivially minimised, completely bypassing the need to identify dimensions that are actually relevant to the underlying taxonomic structure.

This caveat could potentially further be amplified with the introduction of strict disjointness constraints. The requirement of binary assignment choices could incentivise the algorithm to actively look for the dataset's null space. If the data contains variables with incredible small variance or unstructured noise, the algorithm must fully commit to this. This risks populating the disjoint subspace entirely with these low-variance variables if we choose to take a similar block relaxation approach.

There are some potential approaches we can take in order to address this issue. Vichi and Kiers (2001) for one, proposed the idea of removing non significant dimensions from the data, for example by eliminating some of the last principal components of very low variance. However, as detailed throughout, low variance does not necessarily imply a lack of information relating to clustering structure and it is therefore important to realise that this comes at the risk losing such information.

4.3.1 Whitening

Whitening is a procedure by which data can be transformed such that all dimensions possess identical, uncorrelated variance and thus preventing the FKM method from exploiting low-variance noise dimensions. To see this, consider a centered data matrix $\mathbf{X} \in \mathbb{R}^{N\times P}$ which is transformed into some $\mathbf{Z}$ such that the covariance matrix $(1/N)\mathbf{Z}^\top\mathbf{Z} = \mathbf{I}_P$. Then optimising the FKM objective over this whitened feature space results in the leftmost term,

$$\mathrm{Tr}(\mathbf{A}^\top\mathbf{Z}^\top\mathbf{Z}\mathbf{A}) = N\,\mathrm{Tr}(\mathbf{A}^\top\mathbf{A}) = N\,\mathrm{Tr}(\mathbf{I}_L) = NL \quad \text{(const.)}$$

due to the orthonormality constraint $\mathbf{A}^\top\mathbf{A} = \mathbf{I}_L$. As mentioned, this means that the algorithm is no longer rewarded for prioritising the minimisation of subspace variance over the maximisation of between-cluster variance. However any whitening transformation satisfies this unit-variance property therefore it is incredibly important to select one that matches our needs. In fact there are infinitely many possible transformations that satisfy this property due to rotational freedom.

For example, PCA whitening aligns the new coordinate axes with the principal components, effectively rotating the feature space. For clustering interpretation, this is highly undesirable, as the resulting whitened space would describe abstract linear combinations rather than the original physical variables. The FKM procedure would further distort this space and carries the risk of additional information loss.

To preserve spatial interpretability, we choose to utilise Zero Phase Component Analysis (ZCA) whitening. This is because the ZCA transformation is the unique whitening transformation that maximises the average cross-covariance between each component of the whitened and original vectors, see Kessy et al. (2018). In essence this ensures that the each of the P features of the whitened space are correlated with the corresponding feature in the original space. This correlation means that the FKM procedure on this space derives loadings that would in effect, describe relationships between lower-dimensional whitened components and the original unwhitened features. The symmetric ZCA whitening matrix is defined as:

$$\mathbf{W}_{ZCA} = \left(\frac{1}{N}\mathbf{X}^\top\mathbf{X}\right)^{-1/2}$$

which can be computed via a spectral decomposition. The whitened dataset can be obtained by setting $\mathbf{Z} = \mathbf{X}\mathbf{W}_{ZCA}$ and this induces a necessary post-processing step for interpretability based on original variables. If $\mathbf{A}_{ZCA}$ is the FKM-optimal loading matrix in the whitened space then we can derive the porjection,

$$\mathbf{Z}\mathbf{A}_{ZCA} = \mathbf{X}\mathbf{W}_{ZCA}\mathbf{A}_{ZCA} = \mathbf{X}(\mathbf{W}_{ZCA}\mathbf{A}_{ZCA})$$

Therefore we can obtain a corresponding projection matrix $\mathbf{A} = \mathbf{W}_{ZCA}\mathbf{A}_{ZCA}$ for the original dataset $\mathbf{X}$. This recovery step allows for the reversal of spatial distortion from the whitening transformation and is enabled by the correlational relationships between the original and whitened features.

While this is highly effective in the standard case, it introduces an additional structural complication if there are strict disjointness constraints on the FKM model. Since the procedure is applied on the whitened space, it means that only the resulting $\mathbf{A}_{ZCA}$ is disjoint and left-multiplying by $\mathbf{W}_{ZCA}$ to obtain the loadings for the original feature space will likely destroy this disjointness. There are some possible considerations we can take to mitigate this issue.

As shown by Eldar and Oppenheim (2003), the symmetric ZCA matrix is the unique whitening transformation that minimises the Frobenius distance to the identity matrix, $\|\mathbf{W} - \mathbf{I}_P\|_F^2$. Therefore, the resulting transformation likely suppresses off-diagonal entries and the final matrix $\mathbf{A}$ likely retains the dominant loadings from the disjoint whitened space. So while it may not be perfectly sparse or disjoint, it may still result in a much more interpretable space than standard FKM. This could also be aided by clipping proportionally smaller loadings by thresholding and renormalising the columns of $\mathbf{A}$.

Alternatively, the transformation matrix can be restricted to a purely diagonal scaling matrix, $\mathbf{W}_{diag} = \text{diag}(\sigma_1^{-1}, \dots, \sigma_P^{-1})$ rather than $\mathbf{W}_{ZCA}$. This preserves the disjointness from the whitened FKM solution, at the cost of only approximating the identity matrix as it does not decorrelate the original variables. This can be especially dangerous when the original space is composed of highly-correlated variables some caution is required.

4.3.2 Factor Discriminant K-means

The structural vulnerabilities of the FKM objectives have been noted by Rocci et al. (2011) who introduce a modification in order to fix the total variance in the reduced space and they name their proposed method 'Factor Discriminant K-means'. They reformulate the clustering problem through the lens of Linear Discriminant Analysis (LDA) and optimise the FKM objective subject to the constraint $\mathbf{A}^\top\mathbf{X}^\top\mathbf{X}\mathbf{A}$ (rather than orthonormality on $\mathbf{A}$) by alternating a step of LDA and a step of K-means. According to the authors, the Factor Discriminant K-means solution can be obtained by fitting FKM or RKM on the standardised principal components of the original data. This is similar to the ideas behind whitening and confirms that the FKM objective can be stabilised by neutralising the variance discrepancies across the feature space. However, because the solution implicitly relies on principal components, it suffers from the interpretability issues discussed previously, rendering it unsuitable for a model that yields disjointness.

4.3.3 Practical Implementations

In contrast to the idea of pre-processing data to minimise exactly the FKM objective, notable implementations of Factorial K-means, namely the `drclust` package by Prunila and Vichi (2025), take a completely different approach. Instead of minimising the within-cluster variance i.e. the FKM objective, their implementation implicitly treats the projected subspace variance $\mathrm{Tr}(\mathbf{A}^\top\mathbf{X}^\top\mathbf{X}\mathbf{A})$, as locally constant during the subspace update. This effectively substitutes the theoretical FKM subspace update with the leading eigenvectors of $\mathbf{C}_w^\top\mathbf{C}_w$ i.e. $\mathbf{X}^\top\mathbf{P}_\mathbf{U}\mathbf{X}$, forcing the model to maximise actual cluster separation and bypassing the vulnerability to low-variance noise.

4.3.4 Alternative sub-space loop

Recall the initial primary objective which was to introduce disjointness constraints onto Factorial K-means and even further, to extend such notions to the Generalised Reduced Clustering model. Assuming an ALS procedure, in terms of the cluster-partition update we have the trivial solution of falling back to K-means and thus the source of potential obstacles remains the update of the loadings matrix $\mathbf{A}$. As such, drawing upon the theoretical consensus established in the previous sections, a suitable way to proceed is by following the steps of Macedo and Freitas (2015); Prunila and Vichi (2025) and implementing an update that focuses on maximising the between-cluster separation $\mathrm{Tr}(\mathbf{A}^\top\mathbf{X}^\top\mathbf{P}_\mathbf{U}\mathbf{X}\mathbf{A})$. This allows for additional freedom in terms of pre-processing as it mirrors the effect of minimising the standard FKM objective with whitened data. Moreover, it allows for performance comparisons between different pre-processing methods themselves and also between unwhitened and whitened data.

While the CDPCA loop (Algorithm 4) essentially achieves this, it requires either SVD or eigenvector computations at each iteration and block which may scale up computationally in higher dimensions. This may lead to larger issues as we anticipate future extensions of this framework onto Generalised Reduced Clustering, which requires cross-validation over parameters. We therefore explore an alternative way of approaching the disjoint subspace update through linear projections, exploiting the proposed disjoint nature of $\mathbf{A}$.

To begin with, recall the weighted centroids matrix $\mathbf{C}_w = (\mathbf{U}^\top\mathbf{U})^{-1/2}\mathbf{U}^\top\mathbf{X}$ and let $\bar{\mathbf{x}}_p$ denote the p-th column of this matrix. Similarly, let $\bar{\mathbf{y}}_l$ be the l-th column of $\mathbf{C}_w\mathbf{A}$, the low-dimensional projection of these centroids. We can split $\mathbf{A} = \mathbf{V} \odot \tilde{\mathbf{A}}$ where $\tilde{\mathbf{A}}$ is an unconstrained coefficient matrix and the binary matrix $\mathbf{V}$ acts as an indicator for whether elements $\tilde{a}_{pl}$ actually appear in the disjoint matrix $\mathbf{A}$, i.e. elementwise we have $a_{pl} = v_{pl}\tilde{a}_{pl}$ by the Hadamard product $\odot$.

Since $\mathbf{C}_w$ itself is independent of $\mathbf{A}$, we have that:

$$\underset{\mathbf{A}}{\operatorname{argmax}} \left\{ \operatorname{Tr}(\mathbf{A}^\top \mathbf{C}_w^\top \mathbf{C}_w \mathbf{A}) \right\} = \underset{\mathbf{A}}{\operatorname{argmin}} \left\{ \operatorname{Tr}(\mathbf{C}_w^\top \mathbf{C}_w) - \operatorname{Tr}(\mathbf{A}^\top \mathbf{C}_w^\top \mathbf{C}_w \mathbf{A}) \right\} \tag{4.4}$$

Now considering the RHS,

$$\begin{aligned}
\operatorname{Tr}(\mathbf{C}_w^\top \mathbf{C}_w) - \operatorname{Tr}(\mathbf{A}^\top \mathbf{C}_w^\top \mathbf{C}_w \mathbf{A}) &= \sum_{p=1}^{P} \bar{\mathbf{x}}_p^\top \bar{\mathbf{x}}_p - \sum_{l=1}^{L} \bar{\mathbf{y}}_l^\top \bar{\mathbf{y}}_l \\
&= \sum_{p=1}^{P} \bar{\mathbf{x}}_p^\top \bar{\mathbf{x}}_p - 2\sum_{l=1}^{L} \bar{\mathbf{y}}_l^\top \bar{\mathbf{y}}_l + \sum_{l=1}^{L} \bar{\mathbf{y}}_l^\top \bar{\mathbf{y}}_l \\
&= \sum_{p=1}^{P} \left(\sum_{l=1}^{L} v_{pl} \right) \bar{\mathbf{x}}_p^\top \bar{\mathbf{x}}_p - 2 \sum_{l=1}^{L} \left(\sum_{p=1}^{P} v_{pl} \tilde{a}_{pl} \bar{\mathbf{x}}_p \right)^\top \bar{\mathbf{y}}_l + \sum_{l=1}^{L} \left(\sum_{p=1}^{P} v_{pl} \tilde{a}_{pl}^2 \right) \bar{\mathbf{y}}_l^\top \bar{\mathbf{y}}_l \\
&= \sum_{p=1}^{P} \sum_{l=1}^{L} v_{pl} \bar{\mathbf{x}}_p^\top \bar{\mathbf{x}}_p - 2 \sum_{p=1}^{P} \sum_{l=1}^{L} v_{pl} \tilde{a}_{pl} \bar{\mathbf{x}}_p^\top \bar{\mathbf{y}}_l + \sum_{p=1}^{P} \sum_{l=1}^{L} v_{pl} \tilde{a}_{pl}^2 \bar{\mathbf{y}}_l^\top \bar{\mathbf{y}}_l \\
&= \sum_{p=1}^{P} \sum_{l=1}^{L} v_{pl} \left\| \bar{\mathbf{x}}_p - \tilde{a}_{pl} \bar{\mathbf{y}}_l \right\|_2^2
\end{aligned}$$

where $\| \cdot \|_2^2$ denotes the squared Euclidean norm. This formulation relies on three key properties of the defined matrices. First, the disjointness constraint (4.1) enforced by $\mathbf{V}$ ensures that $\sum_{l=1}^{L} v_{pl} = 1$. Second, by the definition of standard matrix multiplication, the low-dimensional projection of the centroids expands to $\bar{\mathbf{y}}_l = \sum_{p=1}^{P} v_{pl} \tilde{a}_{pl} \bar{\mathbf{x}}_p$. Finally, the orthonormality constraint $\mathbf{A}^\top \mathbf{A} = \mathbf{I}_L$ requires that the columns satisfy $\sum_{p=1}^{P} v_{pl} \tilde{a}_{pl}^2 = 1$.

As a result for the relevant subspace spanned by $\mathbf{A}$, it suffices to find:

$$\underset{\tilde{\mathbf{A}}, \mathbf{V}}{\operatorname{argmin}} \sum_{p=1}^{P} \sum_{l=1}^{L} v_{pl} \left\| \bar{\mathbf{x}}_p - \tilde{a}_{pl} \bar{\mathbf{y}}_l \right\|_2^2$$

Taking partial derivatives with respect to a single weight $\tilde{a}_{pl}$ and equating it to zero, we can derive the optimal coefficients,

$$-2 v_{pl} \bar{\mathbf{y}}_l^\top (\bar{\mathbf{x}}_p - \tilde{a}_{pl} \bar{\mathbf{y}}_l) = 0 \implies \tilde{a}_{pl} = \frac{\bar{\mathbf{y}}_l^\top \bar{\mathbf{x}}_p}{\bar{\mathbf{y}}_l^\top \bar{\mathbf{y}}_l}$$

It is easy to compute these projections for every variable p against every latent factor l to obtain all weight coefficients $\tilde{a}_{pl}$. We then know that for a single p, there is exactly one l such that $v_{pl} = 1$. Therefore, we can use previously explored ideas relating to block relaxation (see de Leeuw (1994)) to see that the optimal $\mathbf{V}$ can be computed for each row p:

$$\mathbf{V}_{p,l} = v_{pl} = \begin{cases} 1 & \text{if } l^* = \operatorname{argmin}_l \|\bar{\mathbf{x}}_p - \tilde{a}_{pl} \bar{\mathbf{y}}_k\|_2^2 \\ 0 & \text{otherwise} \end{cases}$$

By embedding this subspace update within an iterative clustering loop, we arrive at an implementation of one run/random start of the *Disjoint Factorial K-means* procedure outlined below.

Algorithm 5: Disjoint Factorial K-means (DFKM)

Input: $\mathbf{X} \in \mathbb{R}^{N \times P}$, the data matrix
K, the number of clusters
L, the target subspace dimension ϵ, the convergence tolerance
`maxiter`, the maximum number of iterations

Output: Optimal partition $\mathbf{U}$, projection matrix $\mathbf{A}$

Initialise cluster partition $\mathbf{U}$ randomly;
Initialise variable partition $\mathbf{V}$ randomly;
Initialise orthonormal loading matrix $\mathbf{A} \in \mathbb{R}^{P \times L}$ randomly;
`// Initialise objective value and iterations`
$\ell_{old} \leftarrow \infty$; `iter` $\leftarrow 1$;
while $|\ell_{old} - \ell_{new}| > \epsilon$ ***and*** `iter` $\leq$ `maxiter` **do**
  `// Fix A, Update U`
  $\mathbf{U} \leftarrow$ Apply K-means update step on $\mathbf{XA}$;
  `// Pre-compute weighted cluster centroids`
  $\mathbf{C}_w \leftarrow (\mathbf{U}^\top \mathbf{U})^{-1/2} \mathbf{U}^\top \mathbf{X}$;
  `// Fix U, Update A and V`
  for *variable* $p = 1$ *to* P **do**
    $\bar{\mathbf{x}}_p \leftarrow p$-th column of $\mathbf{C}_w$;
    for *component* l*=1 to* L **do**
      $\bar{\mathbf{y}}_l \leftarrow l$-th column of $\mathbf{C}_w \mathbf{A}$;
      $\tilde{a}_{pl} \leftarrow \frac{\bar{\mathbf{y}}_l^\top \bar{\mathbf{x}}_p}{\bar{\mathbf{y}}_l^\top \bar{\mathbf{y}}_l}$;
    end
    $l^* \leftarrow \text{argmin}_{1 \leq l \leq L} \|\bar{\mathbf{x}}_p - \tilde{a}_{pl} \bar{\mathbf{y}}_l\|_2^2$;
    $v_{pl^*} \leftarrow 1$;
    $v_{pm} \leftarrow 0 \quad$ for $m \neq l^*$;
  end
  `// Update A, normalise columns`
  $\mathbf{A} \leftarrow \mathbf{V} \odot \tilde{\mathbf{A}}$;
  for *component* $l = 1$ *to* L **do**
    $\mathbf{A}_{\bullet,l} \leftarrow \mathbf{A}_{\bullet,l} / \|\mathbf{A}_{\bullet,l}\|_2$
  end
  `// Update objective values and iteration`
  $\ell_{old} \leftarrow \ell_{new}$;
  $\ell_{new} \leftarrow \ell_{DFKM}(\mathbf{A}, \mathbf{U})$;
  `iter` $\leftarrow$ `iter` $+ 1$;
end
return $\mathbf{U}, \mathbf{A}$

With the formulation of Algorithm 5, we have successfully established a computationally efficient framework for Disjoint Factorial K-means. By replacing iterative eigenvalue decompositions with a scalar projections, this architecture is highly scalable and provides a base for extending these disjoint principles to more complex frameworks.

4.4 Disjoint Generalised Reduced Clustering (DGRC)

Considering the theoretical concepts introduced on Generalised Reuced Clustering in Section 3.3, particularly regarding the splitting of the feature space into relevant and disturbing variables, we recall the GRC objective:

$$\ell_{GRC}(\mathbf{A},\mathbf{U};\,L_D,\rho_1,\rho_2) = \|\mathbf{X}-\mathbf{X}\mathbf{A}\mathbf{A}^\top\|_F^2 + \rho_1\|\mathbf{X}\mathbf{A}_\mathbf{C}-\mathbf{P}_\mathbf{U}\mathbf{X}\mathbf{A}_\mathbf{C}\|_F^2 + \rho_2\|\mathbf{P}_\mathbf{U}\mathbf{X}\mathbf{A}_\mathbf{C}\|_F^2$$

This leads to the proposed constrained *Disjoint Generalised Reduced Clustering* objective:

$$\ell_{DGRC}(\mathbf{A},\mathbf{U};\,L_D,\rho_1,\rho_2) = \ell_{GRC}(\mathbf{A},\mathbf{U};\,L_D,\rho_1,\rho_2) \quad \text{subject to constraints 4.1 on } \mathbf{A}_C \tag{4.5}$$

In particular, note that the constraints are only set on $\mathbf{A}_C$ rather than the entire matrix $\mathbf{A}$ since only the former is relevant in the context of interpretation and this allows us to avoid additional and unnecessary computational complexity.
The subspace update based on gradient projection proposed by Yamamoto and Hwang (2014) operates on a continuous orthogonal manifold, which makes enforcing a strict, discrete disjointness incredibly difficult. Because gradient-based methods are designed for continuous spaces, trying to force a discrete and binary structure onto the loading matrix is both computationally expensive and can even lead to the losing guarantees in terms of minimising the objective. This motivates an alternative approach by reformulating the ℓ_{DGRC} objective for the minimisation with respect to the subspace spanned by $\mathbf{A} = \big(\mathbf{A}_C|\mathbf{A}_D\big)$. Expanding the objective we can derive,

$$\mathrm{Tr}(\mathbf{X}^\top\mathbf{X}) - \mathrm{Tr}(\mathbf{A}^\top\mathbf{X}^\top\mathbf{X}\mathbf{A}) + \rho_1\left(\mathrm{Tr}(\mathbf{A}_C^\top\mathbf{X}^\top\mathbf{X}\mathbf{A}_C) - \mathrm{Tr}(\mathbf{A}_C^\top\mathbf{X}^\top\mathbf{P}_\mathbf{U}\mathbf{X}\mathbf{A}_C)\right) + \rho_2\,\mathrm{Tr}(\mathbf{A}_C^\top\mathbf{X}^\top\mathbf{P}_\mathbf{U}\mathbf{X}\mathbf{A}_C)$$

Using the fact that $\mathbf{A}^\top\mathbf{A} = \mathbf{I}_L$, we must also have that $\mathbf{A}_C^\top\mathbf{A}_D = \mathbf{0}_{(L_C\times L_D)}$ and $\mathbf{A}_D^\top\mathbf{A}_C = \mathbf{0}_{(L_D\times L_C)}$, i.e. $\mathbf{A}_C$ and $\mathbf{A}_D$ are orthogonal to each other and thus we can split up,

$$\mathrm{Tr}(\mathbf{A}^\top\mathbf{X}^\top\mathbf{X}\mathbf{A}) = \mathrm{Tr}(\mathbf{A}_C^\top\mathbf{X}^\top\mathbf{X}\mathbf{A}_C) + \mathrm{Tr}(\mathbf{A}_D^\top\mathbf{X}^\top\mathbf{X}\mathbf{A}_D)$$

We can now further apply block relaxation ideas for the task of minimising the objective over $\mathbf{A}$ and consider independently minimising 4.5 over $\mathbf{A}_C$ and $\mathbf{A}_D$. Collecting such relevant terms,

$$\begin{aligned}
\underset{\mathbf{A}_C}{\mathrm{argmin}}\,\ell_{DGRC}(\mathbf{A},\mathbf{U};\,L_D,\rho_1,\rho_2) &= \underset{\mathbf{A}_C}{\mathrm{argmin}}\left\{(\rho_1-1)\,\mathrm{Tr}(\mathbf{A}_C^\top\mathbf{X}^\top\mathbf{X}\mathbf{A}_C) + (\rho_2-\rho_1)\,\mathrm{Tr}(\mathbf{A}_C^\top\mathbf{X}^\top\mathbf{P}_\mathbf{U}\mathbf{X}\mathbf{A}_C)\right\} \\
&= \underset{\mathbf{A}_C}{\mathrm{argmax}}\left\{(1-\rho_1)\,\mathrm{Tr}(\mathbf{A}_C^\top\mathbf{X}^\top\mathbf{X}\mathbf{A}_C) + (\rho_1-\rho_2)\,\mathrm{Tr}(\mathbf{A}_C^\top\mathbf{X}^\top\mathbf{P}_\mathbf{U}\mathbf{X}\mathbf{A}_C)\right\} \\
&= \underset{\mathbf{A}_C}{\mathrm{argmax}}\,\mathrm{Tr}(\mathbf{A}_C^\top\left[(1-\rho_1)\mathbf{X}^\top\mathbf{X} + (\rho_1-\rho_2)\mathbf{X}^\top\mathbf{P}_\mathbf{U}\mathbf{X}\right]\mathbf{A}_C) \\
&= \underset{\mathbf{A}_C}{\mathrm{argmax}}\,\mathrm{Tr}(\mathbf{A}_C^\top\mathbf{M}\mathbf{A}_C)
\end{aligned}$$

Where we have defined $\mathbf{M}$ as:

$$\mathbf{M} = (1-\rho_1)\mathbf{X}^\top\mathbf{X} + (\rho_1-\rho_2)\mathbf{X}^\top\mathbf{P}_\mathbf{U}\mathbf{X}$$

Remarkably, we have arrived at a form that matches the reformulated DFKM expression 4.4, with $\mathbf{M}$ rather than $\mathbf{C}_w^\top\mathbf{C}_w$. Therefore, it follows that we can also derive discrete updates for the variable partition $\mathbf{V}$ and the clustering-relevant loadings matrix $\mathbf{A}_C$ via linear scalar projections. Given the previously established requirement of $\rho_1 > \rho_2$, it is clear that $\mathbf{M}$ is positive definite whenever $\rho_1 \leq 1$ and in this case such solutions are well-defined.
However it is critical to consider the case where $\rho_1 > 1$, which results in $\mathbf{M}$ no longer being positive definite and the maximisation problem is ill-posed.

In order to circumvent this issue and ensure robustness for any parameterisation of ρ_1, we introduce a shifted $\mathbf{M}_\epsilon$ whenever $\rho_1 > 1$. To do this pick $\epsilon \geq |\lambda_{min}(\mathbf{M})|$, the smallest (negative) eigenvalue of $\mathbf{M}$ and define:

$$\mathbf{M}_\epsilon := \mathbf{M} + \epsilon\, \mathbf{I}_P$$

This shift ensures that all eigenvalues are strictly positive and hence, $\mathbf{M}_\epsilon$ is positive definite and we can fall back to the updates introduced for DFKM. Moreover,

$$\mathrm{Tr}(\mathbf{A}_C^\top \mathbf{M}_\epsilon \mathbf{A}_C) = \mathrm{Tr}(\mathbf{A}_C^\top \mathbf{M} \mathbf{A}_C) + \epsilon\, \mathrm{Tr}(\mathbf{A}_C^\top \mathbf{I}_P \mathbf{A}_C) = \mathrm{Tr}(\mathbf{A}_C^\top \mathbf{M} \mathbf{A}_C) + \epsilon L_C$$

Since ϵL_C is a constant, the optimal $\mathbf{A}_C$ is independent of this i.e.

$$\underset{\mathbf{A}_C}{\mathrm{argmax}}\, \mathrm{Tr}(\mathbf{A}_C^\top \mathbf{M}_\epsilon \mathbf{A}_C) = \underset{\mathbf{A}_C}{\mathrm{argmax}}\, \mathrm{Tr}(\mathbf{A}_C^\top \mathbf{M} \mathbf{A}_C)$$

Without loss of generality, we drop this ϵ and consider abstractly a positive definite symmetric matrix $\mathbf{M}$ and the above maximisation problem. As we did for DFKM, if we define the form $\mathbf{A}_C = \mathbf{V} \odot \tilde{\mathbf{A}}_C$, we can derive the following optimal coefficient updates:

$$\tilde{a}_{pl} = \frac{\mathbf{A}_{C_{\bullet,l}}^\top \mathbf{M}_{\bullet,p}}{\mathbf{A}_{C_{\bullet,l}}^\top \mathbf{M} \mathbf{A}_{C_{\bullet,l}}}$$

The form of the denominator gives some intuition behind the need for positive definiteness. The $\mathbf{V}$ updates can then be set up exactly the same way as seen in subsection 4.3.4.
All that remains to be considered is the optimal choice for the disturbing subspace sppanned by $\mathbf{A}_D$. Recalling the expansion of the DGRC objective, the independent minimisation problem over $\mathbf{A}_D$ terms is:

$$\underset{\mathbf{A}_D}{\mathrm{argmin}}\, \ell_{DGRC}(\mathbf{A}, \mathbf{U};\, L_D, \rho_1, \rho_2) = \underset{\mathbf{A}_D}{\mathrm{argmax}}\, \mathrm{Tr}(\mathbf{A}_D^\top \mathbf{X}^\top \mathbf{X} \mathbf{A}_D)$$

$$\text{subject to} \quad \mathbf{A}_D^\top \mathbf{A}_D = \mathbf{I}_{L_D} \quad \text{and} \quad \mathbf{A}_C^\top \mathbf{A}_D = \mathbf{0}$$

The subspace spanned by $\mathbf{A}_D$ is tasked with capturing the residual variance that does not contribute to the clustering structure. Although it is not bound by the discrete disjointness constraints imposed on $\mathbf{A}_C$, it must reside within the orthogonal complement of $\mathbf{A}_C$. To enforce this constraint, we can use the null-space projection matrix of $\mathbf{A}_C$ which is $\mathbf{I}_P - \mathbf{A}_C \mathbf{A}_C^\top$ and apply it onto the total variance term to obtain the form:

$$\left(\mathbf{I}_P - \mathbf{A}_C \mathbf{A}_C^\top\right)^\top \mathbf{X}^\top \mathbf{X} \left(\mathbf{I}_P - \mathbf{A}_C \mathbf{A}_C^\top\right)$$

By substituting this into the $\mathbf{A}_D$ minimisation problem, we can effectively remove the orthogonality constraint against $\mathbf{A}_C$ and derive the problem:

$$\underset{\mathbf{A}_D}{\mathrm{argmax}}\, \mathrm{Tr}(\mathbf{A}_D^\top \left(\mathbf{I}_P - \mathbf{A}_C \mathbf{A}_C^\top\right)^\top \mathbf{X}^\top \mathbf{X} \left(\mathbf{I}_P - \mathbf{A}_C \mathbf{A}_C^\top\right) \mathbf{A}_D) \quad \text{subject to} \quad \mathbf{A}_D^\top \mathbf{A}_D = \mathbf{I}_{L_D}$$

This problem has an exact solution (which we have come across many times). We conclude that the optimal disturbing loading matrix $\mathbf{A}_D$ is formed by extracting the L_D leading right-singular vectors of $\mathbf{X}\left(\mathbf{I}_P - \mathbf{A}_C \mathbf{A}_C^\top\right)$.

Algorithm 6: Disjoint Generalised Reduced Clustering (DGRC)

Input: $\mathbf{X} \in \mathbb{R}^{N \times P}$, the data matrix
K, the number of clusters
L_C, the target dimension of the relevant subspace
L_D, the target dimension of the disturbing subspace
ρ_1, ρ_2, the Generalised Reduced Clustering weighting parameters
ϵ_{tol}, the convergence tolerance
`maxiter`, the maximum number of iterations

Output: Optimal partition $\mathbf{U}$, projection matrix $\mathbf{A} = \big(\mathbf{A}_C | \mathbf{A}_D\big)$

Initialise cluster partition $\mathbf{U}$ randomly;
Initialise variable partition $\mathbf{V}$ randomly;
Initialise orthonormal relevant loading matrix $\mathbf{A}_C \in \mathbb{R}^{P \times L_C}$ randomly;
`// Initialise objective value and iterations`
$\ell_{old} \leftarrow \infty$; `iter` $\leftarrow 1$;
while $|\ell_{old} - \ell_{new}| > \epsilon_{tol}$ ***and*** `iter` $\leq$ `maxiter` **do**
    `// Fix A, Update U`
    $\mathbf{U} \leftarrow$ Apply K-means update step on $\mathbf{X}\mathbf{A}_C$;
    `// Construct M and apply shift if necessary`
    $\mathbf{M} \leftarrow (1-\rho_1)\mathbf{X}^\top\mathbf{X} + (\rho_1 - \rho_2)\mathbf{X}^\top\mathbf{P}_\mathbf{U}\mathbf{X}$;
    $\epsilon \leftarrow |\lambda_{min}(\mathbf{M})|$ **if** $\rho_1 > 1$ **else** 0;
    $\mathbf{M} \leftarrow \mathbf{M} + \epsilon\mathbf{I}_P$;
    `// Fix U, Update Ac and V`
    for *variable* $p = 1$ *to* P **do**
        for *component* $l = 1$ *to* L_C **do**
            $\tilde{a}_{pl} \leftarrow (\mathbf{A}_{C_{\bullet,l}}^\top \mathbf{M}_{\bullet,p}) / (\mathbf{A}_{C_{\bullet,l}}^\top \mathbf{M}\mathbf{A}_{C_{\bullet,l}})$;
        end
        $l^* \leftarrow \operatorname{argmin}_{1 \leq l \leq L_C} \Big(\mathbf{M}_{p,p} - 2\tilde{a}_{pl}(\mathbf{A}_{C_{\bullet,l}}^\top \mathbf{M}_{\bullet,p}) + \tilde{a}_{pl}^2(\mathbf{A}_{C_{\bullet,l}}^\top \mathbf{M}\mathbf{A}_{C_{\bullet,l}})\Big)$;
        $v_{pl^*} \leftarrow 1$;
        $v_{pm} \leftarrow 0 \quad \text{for } m \neq l^*$;
    end
    `// Update Ac, normalise columns`
    $\mathbf{A}_C \leftarrow \mathbf{V} \odot \tilde{\mathbf{A}}_C$;
    for *component* $l = 1$ *to* L_C **do**
        $\mathbf{A}_{C_{\bullet,l}} \leftarrow \mathbf{A}_{C_{\bullet,l}} / \|\mathbf{A}_{C_{\bullet,l}}\|_2$;
    end
    `// Extract orthogonal disturbing subspace`
    if $L_D > 0$ **then**
        $\mathbf{A}_D \leftarrow L_D$ leading right-singular vectors of $\mathbf{X}(\mathbf{I}_P - \mathbf{A}_C\mathbf{A}_C^\top)$;
    end
    $\mathbf{A} \leftarrow \big(\mathbf{A}_C | \mathbf{A}_D\big)$;
    `// Update objective values and iteration`
    $\ell_{old} \leftarrow \ell_{new}$;
    $\ell_{new} \leftarrow \ell_{DGRC}(\mathbf{A}, \mathbf{U}; L_D, \rho_1, \rho_2)$;
    `iter` $\leftarrow$ `iter` $+ 1$;
end
return $\mathbf{U}, \mathbf{A}$

Chapter 5

Methods & Results

Throughout this chapter, we present an empirical evaluation of the Disjoint Generalised Reduced Clustering (DGRC) algorithm, systematically benchmarking its ability to recover accurate cluster partitions and its robustness against various noise configurations, in comparison to the established simultaneous dimensionality reduction and clustering methods detailed in the preceding chapters.

To facilitate a rigorous and unified comparative analysis, all algorithmic procedures and baseline models were implemented in Python. The baseline algorithms Reduced K-means (RKM), Factorial K-means (FKM), and Cluster Disjoint Principal Component Analysis (CDPCA) were translated from their implementations in `R` and `C++` by Prunila and Vichi (2025). The Generalised Reduced Clustering (GRC) algorithm was obtained by translating the corresponding `C` implementation by Yamamoto (2014). The details of these are implementations are clarified in the Appendix Section A.1. The DGRC algorithm is implemented originally in Python and the source code is available in a repository linked in Appendix A.1.

Before detailing the experimental results, we first establish the theoretical frameworks required to formally measure clustering partition accuracy and to systematically control subspace variance and cluster ambiguity within synthetically generated data. We then aim to explore the usage of DGRC on real world data.

To quantify the accuracy of recovered cluster assignments against assumed latent partitions, we can employ the Adjusted Rand Index (ARI) Hubert and Arabie (1985). Let $\mathbf{U} = \{u_1, \ldots, u_K\}$ denote the true latent partition of N observations, and $\hat{\mathbf{U}} = \{\hat{u}_1, \ldots, \hat{u}_K\}$ denote a partition recovered by a clustering algorithm. Their intersection forms a contingency table where each entry $n_{ij} = |u_i \cap \hat{u}_j|$ denotes the number of observations simultaneously placed in true cluster u_i and recovered cluster $\hat{u}_j$. Let $a_i = \sum_{j=1}^{K} n_{ij}$ and $b_j = \sum_{i=1}^{K} n_{ij}$ represent their respective sums. The ARI evaluates the proportion of observation pairs identically grouped in both $\mathbf{U}$ and $\hat{\mathbf{U}}$, normalising for the expected number of matches under purely random partitioning. The ARI is formally defined as:

$$\text{ARI} = \frac{\sum_{i,j} \binom{n_{ij}}{2} - \left[\sum_i \binom{a_i}{2} \sum_j \binom{b_j}{2}\right] / \binom{N}{2}}{\frac{1}{2}\left[\sum_i \binom{a_i}{2} + \sum_j \binom{b_j}{2}\right] - \left[\sum_i \binom{a_i}{2} \sum_j \binom{b_j}{2}\right] / \binom{N}{2}} \tag{5.1}$$

5.1 Simulation Study

We aim to evaluate the capacity of DGRC to recover the correct subspace and the correct cluster memberships in comparison to RKM, FKM and CDPCA under various controlled conditions. RKM and FKM serve as theoretical baselines which each possess their own vulnerabilities whereas CDPCA represents an alternative disjoint framework. In order to achieve a comprehensive set of comparisons, we conduct a simulation study under different levels of subspace variance and orthogonal residual variance. We explore methods by which these factors can be controlled, leaning on ideas introduced by Timmerman et al. (2010) in comparing the performances of RKM and FKM and the evaluation of GRC by Yamamoto and Hwang (2014).

Following the notion of subspace separation explored in Section 3.3, we construct a $N \times P$ synthetic simulation dataset $\mathbf{X}_{\text{sim}}$ assuming the decomposition:

$$\mathbf{X}_{\text{sim}} = \mathbf{P}_{\mathbf{U}}\mathbf{X}\mathbf{A}_C\mathbf{A}_C^\top + \mathbf{E}_C\mathbf{A}_C^\top + \mathbf{E}_D\mathbf{A}_D^\top + \mathbf{E}^\perp \tag{5.2}$$

The $\mathbf{P}_{\mathbf{U}}\mathbf{X}\mathbf{A}_C\mathbf{A}_C^\top$ term has been explored before and it represents the reconstruction of the K latent cluster centroids from the reduced space. Computationally, this term is generated by initialising K equidistant L_C-dimensional centroids and a true cluster membership matrix $\mathbf{U}$. We fix the expected proportion of objects in $\mathbf{U}$ to be equal across clusters, while L_C is set to $K-1$ as recommended by Vichi and Kiers (2001). The $\mathbf{E}_C\mathbf{A}_C^\top$ represents the clustering-relevant residuals and projects the within-cluster variance onto the P-dimensional dataset. The residuals are sampled independently $\mathbf{E}_C \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_{L_C})$. To maintain consitency, the $P \times L_C$ loading matrix $\mathbf{A}_C$ is constructed to obey the disjointness constraints. It is crucial to note that we set some $P_1 \leq P$ such that only the first P_1 rows of $\mathbf{A}_C$ contain non-zero loadings which means that the L_C clustering relevant latent components only influence the first P_1 variables of $\mathbf{X}_{\text{sim}}$.

Similarly, the $\mathbf{E}_D\mathbf{A}_D^\top$ term consists of L_D-dimensional noise generated by $\mathbf{E}_D \sim \mathcal{N}(\mathbf{0}, \sigma_D^2\mathbf{I}_{L_D})$ where σ_D is configured in a way that the variance in the L_D-dimensional subspace is equal to the variance in the L_C-dimensional cluster space. This matrix $\mathbf{E}_D$ is then projected onto the observed space through the transpose of the $P \times L_D$ loading matrix $\mathbf{A}_D$. This loading matrix is not under any disjointness constraints and is generated by setting some $P_2 \leq P$ and sampling $P_2 \times L_D$ independent standard Gaussian loadings. $\mathbf{A}_D$ is then constructed by only filling the $(P_1 + 1)$ through $P_1 + P_2$ rows with loadings so that the L_D noise components only influence the P_2 variables of $\mathbf{X}_{\text{sim}}$ after P_1.

Finally the $\mathbf{E}^\perp$ term represents the unstructured background noise spanning the remainder of the high-dimensional ambient space. To prevent this noise from artificially inflating the subspace residuals, a $N \times P$ matrix of Gaussian noise $\eta \sim \mathcal{N}(\mathbf{0}, \sigma_\perp^2\mathbf{I}_P)$ is generated and projected onto the null-space of the entire loading matrix $\mathbf{A} = (\mathbf{A}_C \mid \mathbf{A}_D)$, by setting $\mathbf{E}^\perp := \eta - \eta\mathbf{A}\mathbf{A}^\top$. The choice of $\sigma_\perp$ is not fixed but a controlled parameter, and its effect and role will be detailed further on. Moreover, this means that the remaining final P_3 variables of $\mathbf{X}_{\text{sim}}$ are devoid of any structural or masking signal, consisting exclusively of the projected ambient residuals originating from $\mathbf{E}^\perp$. Overall, by partitioning the feature space such that $P = P_1 + P_2 + P_3$, we implicitly decouple the relevant clustering information from the correlated masking variables, and the ambient noise.

To visualise this decomposition and the resulting variable assignments, consider a simplified configuration where the observed space $P = 6$ is partitioned into $P_1 = 2$, $P_2 = 2$, and $P_3 = 2$, with single latent dimensions $L_C = 1$ and $L_D = 1$. The transposed loading matrices determine which observed variables receive which latent signals so consider:

$$\mathbf{A}_C^\top = \begin{bmatrix} a_{C,1} & a_{C,2} & 0 & 0 & 0 & 0 \end{bmatrix}, \quad \mathbf{A}_D^\top = \begin{bmatrix} 0 & 0 & a_{D,1} & a_{D,2} & 0 & 0 \end{bmatrix}$$

Then for the simulated dataset, denoting the centroids $\bar{\mathbf{y}} = \mathbf{P}_{\mathbf{U}}\mathbf{X}\mathbf{A}_C$, the model expands to:

$$\begin{aligned}
\mathbf{X}_{\text{sim}} &= (\mathbf{P}_{\mathbf{U}}\mathbf{X}\mathbf{A}_C + \mathbf{E}_C)\mathbf{A}_C^\top + \mathbf{E}_D\mathbf{A}_D^\top + \mathbf{E}^\perp \\
&= (\bar{\mathbf{y}} + \mathbf{e}_C)\begin{bmatrix} a_{C,1} & a_{C,2} & 0 & 0 & 0 & 0 \end{bmatrix} + \mathbf{e}_D\begin{bmatrix} 0 & 0 & a_{D,1} & a_{D,2} & 0 & 0 \end{bmatrix} + \begin{bmatrix} \mathbf{e}_1^\perp & \mathbf{e}_2^\perp & \mathbf{e}_3^\perp & \mathbf{e}_4^\perp & \mathbf{e}_5^\perp & \mathbf{e}_6^\perp \end{bmatrix} \\
&= \Big[\, \underbrace{(\bar{\mathbf{y}} + \mathbf{e}_C)a_{C,1} + \mathbf{e}_1^\perp \quad (\bar{\mathbf{y}} + \mathbf{e}_C)a_{C,2} + \mathbf{e}_2^\perp}_{P_1} \quad \underbrace{\mathbf{e}_D a_{D,1} + \mathbf{e}_3^\perp \quad \mathbf{e}_D a_{D,2} + \mathbf{e}_4^\perp}_{P_2} \quad \underbrace{\mathbf{e}_5^\perp \quad \mathbf{e}_6^\perp}_{P_3} \,\Big]
\end{aligned}$$

where $\mathbf{e}_C$, $\mathbf{e}_D$, and each $\mathbf{e}_j^\perp$ are $N \times 1$ residual terms.

5.1.1 Controlled Parameters

In generating these synthetic datasets, it is important to clarify both the assumed geometric structure of the data and the experimental conditions we wish to evaluate. For the scope of this simulation study, we do not vary the number of clusters K, the latent clustering dimensionality L_C, the disturbing dimensionality L_D, or the spatial separation/position between cluster centroids. Instead, these are held constant as fixed latent assumptions to ensure a stable foundation across all trials.

What we control and use to evaluate algorithmic performance are the observable dimensional parameters: N, P_1, P_2, and P_3. These parameters dictate the scale of the high-dimensional space and represent the number of observed variables corresponding to each respective latent block. By varying P_1, P_2, and P_3, we can directly evaluate the robustness of the models as the number of irrelevant dimensions in the feature space scales up.

Beyond the dimensionality, the second axis of evaluation relies on the notion of Proportion of Subspace Residuals (PSR), as seen in Timmerman et al. (2010), which represents the proportion of total dataset variance attributed to the relevant clustering structure. Recall that the internal cluster residuals $\mathbf{E}_C$ are generated using standard Gaussians with fixed unit variance. Similarly, the disturbing subspace residuals $\mathbf{E}_D$ are calibrated such that their variance exactly matches that of the relevant clustering subspace. In contrast, the remaining ambient noise $\mathbf{E}^{\perp}$ is scaled by the variance parameter $\sigma_{\perp}$ and therefore we can directly control the proportion of the total observed variance explained by the true clustering structure by varying $\sigma_{\perp}$. This parameterisation allows us to test the algorithms under varying levels of proportional background noise.

5.1.2 Evaluation

The latent structural dimensions are held constant as mentioned above, and in particular we set global conditions $K = 4$, $L_C = 3$, and $L_D = 2$ across all simulated datasets. Utilising our control over N, P_1, P_2, P_3, and PSR (implicitly via $\sigma_{\perp}$), we construct two primary streams of analysis to evaluate the performances of RKM, FKM, CDPCA, and DGRC under various conditions and controlled parameter configurations. For every unique parameter configuration, we generate 80 independent datasets where accurate recovery of cluster memberships measured using the Adjusted Rand Index (ARI) . For each unique configuration of (P_1, P_2, P_3 PSR) the DGRC algorithm undergoes a cross-validation procedure to obtain optimal values for the hyperparameters ρ_1 and ρ_2. We validate over $\rho_1 \in \{0.5, 1.0, 1.5, 2.0, 2.5, 3.0\}$ and $\rho_2 \in \{0.0, 0.5, 1.0, 1.5, 2.0\}$ using three validation folds and 15 random initialisations.

The first set of experiments aims to assess the performance of the four clustering procedures under different levels of Proportion of Subspace Residuals (PSR). To achieve this, we restrict the dimensional parameters to be fixed at $P_1 = 6$, $P_2 = 6$ and $P_3 = 12$, while the number of objects is set to $N = 200$. The PSR parameter is assigned values across the set $\{0.001, 0.01, 0.02, 0.03, 0.045, 0.06, 0.08, 0.1\}$. At each level of PSR, the distribution of ARI values for RKM, FKM, CDPCA, and DGRC across 80 independent datasets is visualised as a box plot, as seen in Figure 5.1.

In particular, it can be seen in Figure 5.1 that at very low PSR levels, FKM performs very well while the other methods show poor recovery. This is likely due to FKM benefiting from the cluster subspace lying in a proportionally low variance subspace, as discussed in Section 4.3. As PSR increases to 0.03, where clustering residuals contribute to 3% of the total variance, DGRC starts to perform well, and we see some very accurate recoveries of cluster memberships. This signifies that with more random starts per run, a better median could be achieved at this threshold. Beyond a PSR of 0.045, meaning beyond 4.5% of explained variance, the boxplots for DGRC have median values converging near an ARI of 1. This shows that for this specific parameter configuration, DGRC performs incredibly well and can accurately isolate the clustering space once this PSR threshold is reached. However, overall, it is slightly disappointing that the recovery of cluster memberships for DGRC follows on par with those of RKM and CDPCA, as both of these alternative methods also exhibit median values near one at similar levels of PSR.

The second set of experiments takes inspiration from Yamamoto and Hwang (2014) and relies on varying the number of objects $N \in \{100, 200, 300\}$, the number of disturbing variables $P_2 \in \{0, 5, 10\}$, and the number of random error variables $P_3 \in \{0, 5\}$ while the size of the relevant clustering subspace is determined implicitly via $P_1 = 25 - P_2 - P_3$. Since this requires variation of three different parameters, we achieve this by first considering the scenario where $P_3 = 0$, followed by a noisier environment where $P_3 = 5$ as displayed in Figure 5.2. Again, for each configuration these boxplots display the distribution of ARI values for RKM, FKM, CDPCA, and DGRC across 80 independent datasets. The resulting plots for the $P_3 = 0$ scenario as seen in Figure A.1 have been placed in Appendix A.2

In both scenarios, it can be seen that increasing the sample size N from 100 to 300 consistently leads to higher median values for all methods. When $N = 100$, recovery is varying with wide boxes across all conditions. In Figure A.1, where $P_3 = 0$, the boxplots for RKM, CDPCA, and DGRC have median values near one and narrow boxes when $N \geq 200$ and $P_2 = 0$. However, as the number of disturbing variables increases to $P_2 = 5$ and $P_2 = 10$, the median values for DGRC and CDPCA decrease and their boxes become wider. Interestingly, RKM recovers cluster memberships better than DGRC and CDPCA under these conditions, maintaining higher median values and narrower boxes as P_2 increases, however the tails of the DGRC boxes remain consistent with the ARIs achieved by RKM. In Figure 5.2, where random error variables $P_3 = 5$ are introduced, the median values decrease further for all methods. Across all conditions, FKM has low median values and upper bounds, showing that it does not recover cluster memberships in the presence of these noise variables. DGRC remains robust for $P_2 = 10$ and $P_3 = 5$ if $N = 300$, showing that with enough data samples, DGRC can still successfully overcome the noise variables in this data setup.

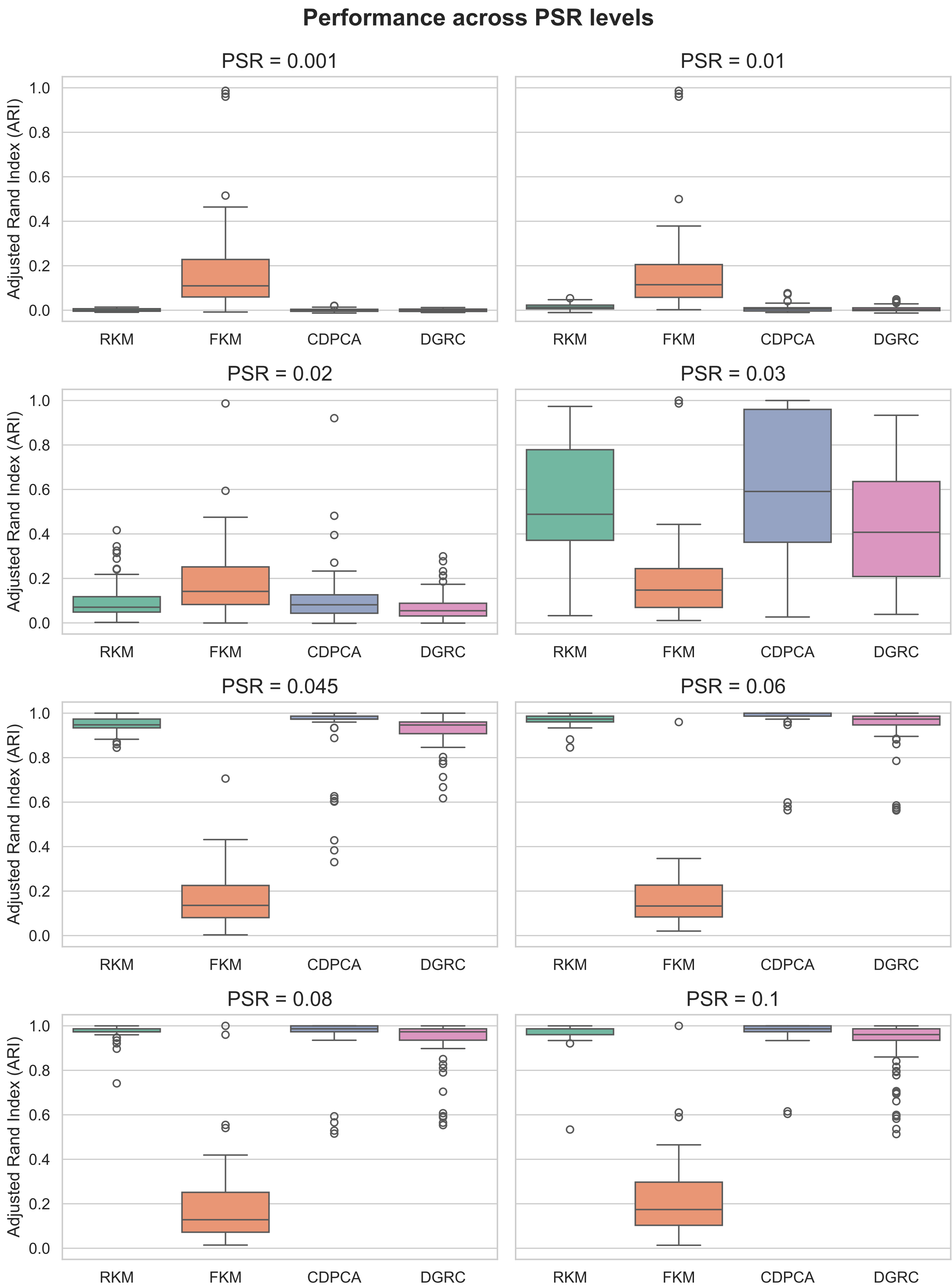

Figure 5.1 Boxplots of Adjusted Rand Indices for RKM, FKM, CDPCA and DGRC over PSR

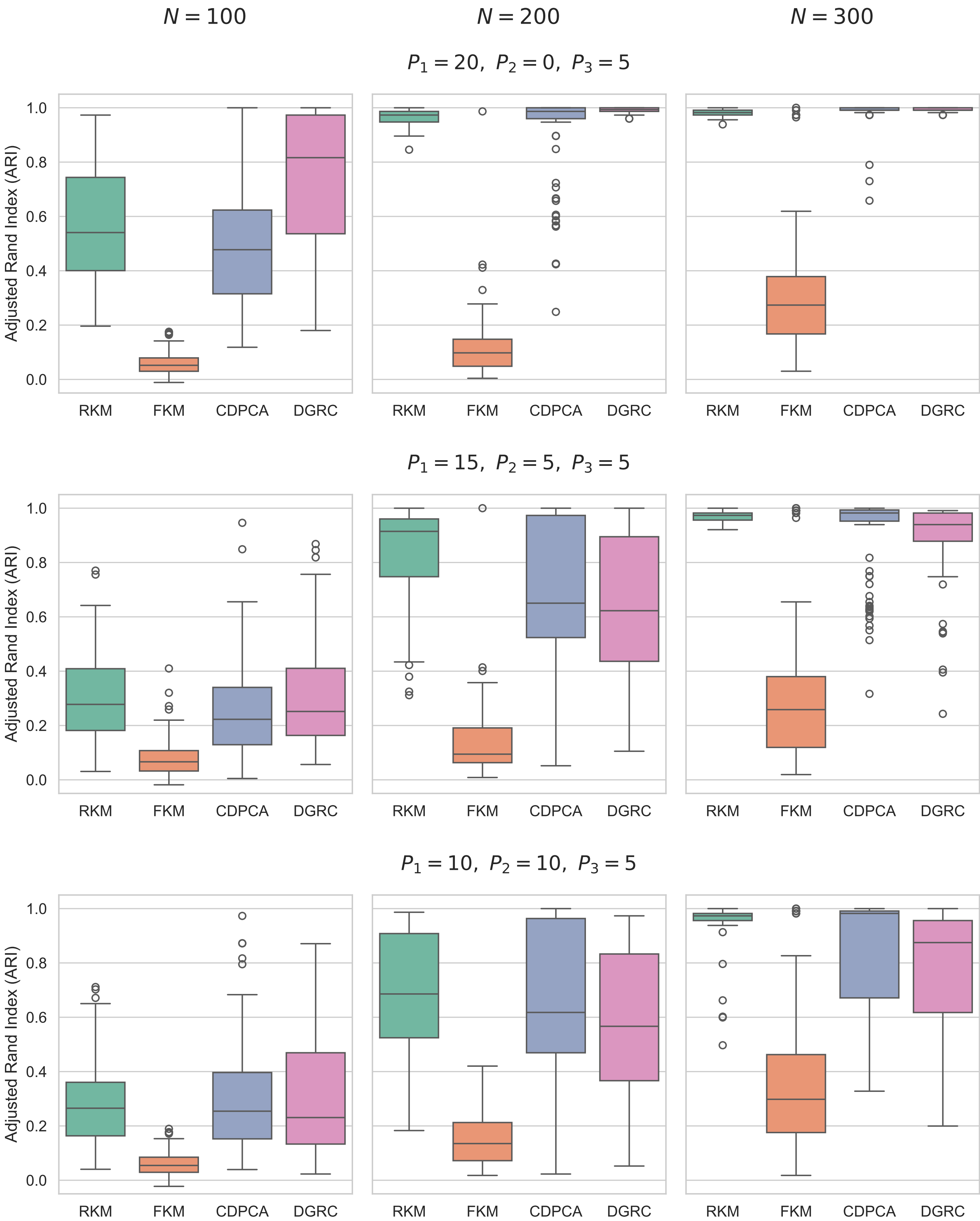

Figure 5.2 Boxplots of Adjusted Rand Indices for RKM, FKM, CDPCA and DGRC across values of N, P_1, P_2 with fixed $P_3 = 5$

5.2 Application: Global Health Indicators

The clustering methodologies are applied to a real dataset to evaluate their ability to identify classes of countries with similar structural health profiles. The health and health-based economic indicators of 105 sovereign nations (Table 5.4) are analysed using data drawn from Our World in Data (OWID) United Nations et al. (2025); UN Inter-agency Group for Child Mortality Estimation et al. (2025); World Health Organization and World Bank (2026); Food and Agriculture Organization of the United Nations (2025); World Health Organization et al. (2025); Eurostat et al. (2026). A five-year observational window (2015–2019) is utilised to capture the most recent observation of each metric.

Countries are evaluated according to 16 indicators/metrics spanning broad categories such as health outcomes, infrastructure, lifestyle risk factors, and macroeconomic factors. The selected variables are shown in Table 5.1. The aim of this study is to identify exactly which of these individual factors and categories are most useful for driving meaningful geographic clustering. To ensure consistency across different metric scales, the dataset is standardised before passing it through the dimensionality reduction and clustering procedures.

Table 5.1 Selected Global Health Metrics and Structural Categories

Category	**Metrics**
Health Outcomes	Life Expectancy, Child Mortality, Incidence of Tuberculosis (SDGs)
Infrastructure & Access	Physicians per 1000 People, Nurses and Midwives per 1000 People, Hospital Beds per 1000 People, Share of Children Immunized (DTP3), Proportion Using Safely Managed Drinking Water
Lifestyle & Risk Factors	Share of Adults Who Smoke, Total Alcohol Consumption per Capita (Litres of Pure Alcohol), Share of Adults Defined as Obese, Prevalence of Undernourishment
Macroeconomic Factors	Population, GDP per Capita (World Bank), Total Healthcare Expenditure (% GDP), Share of Out-of-Pocket Expenditure on Healthcare

Across all evaluated methodologies, the number of clusters is set to $K = 4$ to align with the United Nations Development Programme's four-tier stratification of global structural capacity UNDP (2025) and human development. Tandem analysis over the first two principal components, alongside a RKM and a preliminary DGRC model ($L_C = 2$), are priorly used for comparative and exploratory purposes. The corresponding projections and component loadings are detailed in Appendix A.3. Because the tandem analysis and RKM result in highly similar plots and loadings for this specific dataset, the primary focus of this analysis is to compare the performances of the disjoint methodologies, specifically CDPCA and DGRC, and their ability to potentially separate these metrics into broader classes.

Cluster Disjoint Principal Component Analysis (CDPCA) is initially executed and the obtained loading matrix is reported in Table 5.2 while the projection onto these components is shown in Figure 5.3. In this instance, CDPCA fails to provide a meaningful, multifaceted clustering structure. It completely isolates the Population variable into the second latent component (with a loading of 1), leaving all other variables grouped entirely in the first component. This is likely driven by the disproportionately large variance of the population metric. As observed in the resulting plot, this strict allocation causes India and China to appear as extreme outliers along the second axis, demonstrating how disjointness constraints can be exploited by a single high-variance variables rather than capturing nuanced structural differences. Moreover, increasing the number of latent dimensions to three did not improve this result and only led to an additional high-variance metric such as Life Expectancy being assigned to the third component.

Table 5.2 CDPCA Loadings for Latent Components 1 and 2

Metric	LC1	LC2
Life Expectancy	0.3302	0
Child Mortality	-0.2996	0
Incidence of Tuberculosis (SDGs)	-0.2374	0
Share of Adults Who Smoke	0.1391	0
Total Alcohol Consumption per Capita (Litres)	0.2082	0
Share of Adults Defined as Obese	0.2392	0
Prevalence of Undernourishment	-0.2332	0
Proportion Using Safely Managed Drinking Water	0.2869	0
Physicians per 1000 People	0.3298	0
Nurses and Midwives per 1000 People	0.2996	0
Hospital Beds per 1000 People	0.2496	0
Share of Children Immunized (DTP3)	0.1885	0
Total Healthcare Expenditure (% GDP)	0.2448	0
Share of Out-of-Pocket Expenditure on Healthcare	-0.1664	0
Population	0	1.0000
GDP per Capita (World Bank)	0.3229	0

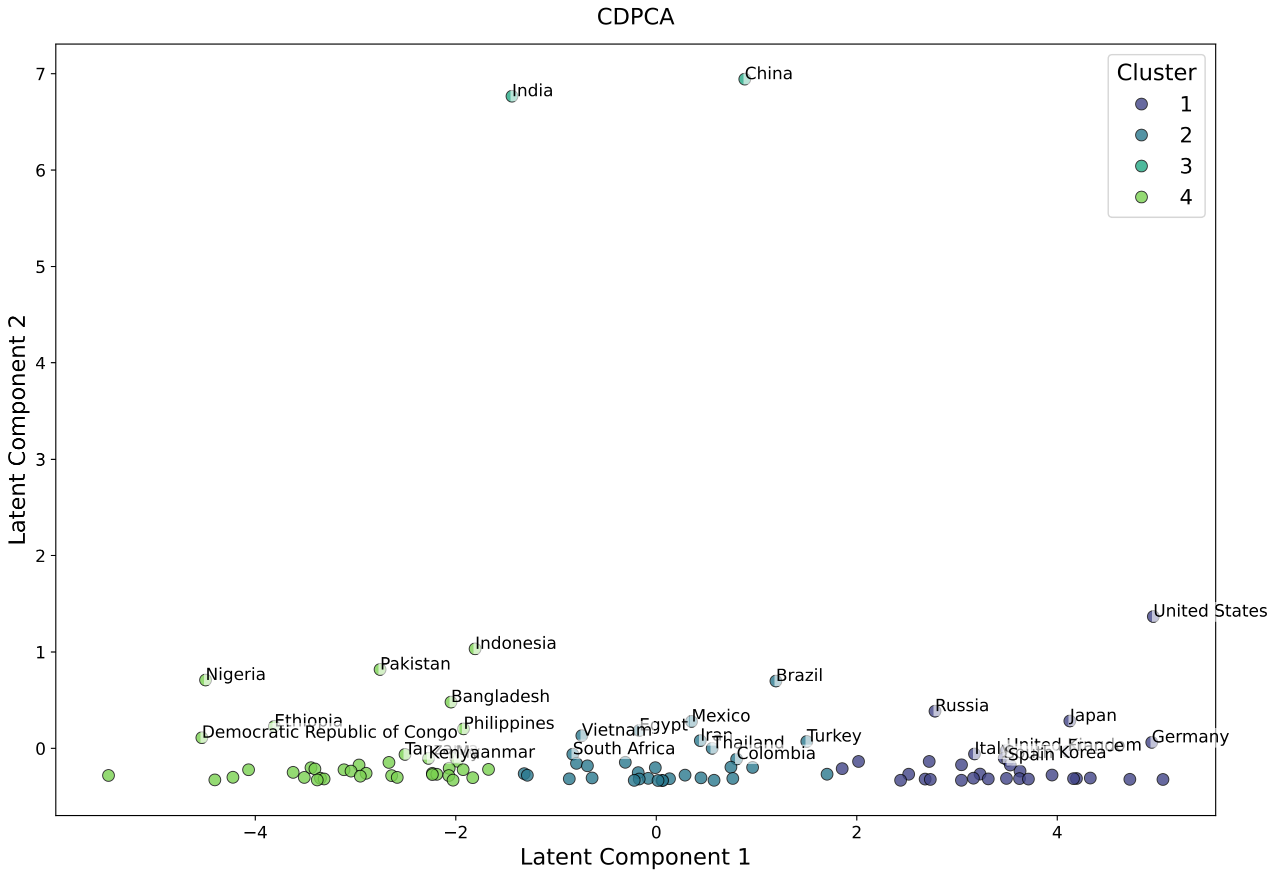

Figure 5.3 Projection onto latent components obtained by CDPCA

In an attempt to address these limitations, the Disjoint Generalised Reduced Clustering (DGRC) procedure is applied. The number of relevant clustering components is set to $L_C = 3$ (which is $K - 1$, as recommended by De Soete and Carroll (1994); Vichi and Kiers (2001)), while the dimensions $L_D = 4$ and hyperparameters $\rho_1 = 3.5, \rho_2 = 0.5$ are obtained through cross-validation. The component loading matrix of the DGRC is reported in Table 5.3. In contrast to CDPCA, DGRC successfully manages to split the broad categories of variables across multiple dimensions. Looking at the relevant subspace, the first latent component (LC1) cleanly isolates foundational health outcomes, namely life expectancy and child mortality. The second component (LC2) captures healthcare financing dynamics, weighting out-of-pocket expenditure heavily alongside specific lifestyle metrics like obesity. The third component (LC3) groups overarching economic capacity (GDP per capita) with vulnerability metrics such as undernourishment and tuberculosis incidence. Meanwhile, the disturbing dimensions (LC4 to LC7) absorb the variance of population size, smoking prevalence, and physical infrastructure. This separation ensures that the core clustering space is defined by differences in metrics related to an underlying clustering based on health-metrics rather than driven by varying metrics.

Table 5.3 DGRC Loadings with $L_C = 3, L_D = 4$

	Relevant			**Disturbing**			
Metric	LC1	LC2	LC3	LC4	LC5	LC6	LC7
Life Expectancy	0.7400	0	0	-0.1452	-0.1741	0.0208	0.3297
Child Mortality	0.6585	0	0	0.1682	0.2313	-0.0606	-0.3209
Incidence of Tuberculosis (SDGs)	0	0	0.2627	0.1678	-0.2234	0.2968	-0.2572
Share of Adults Who Smoke	0.0502	0	0	0.3449	-0.5187	0.4870	0.1541
Total Alcohol Consumption per Capita (Litres)	0	0	0.2954	0.2969	0.3470	0.1321	0.1781
Share of Adults Defined as Obese	0	0.4017	0	-0.0404	-0.4369	-0.3549	0.2401
Prevalence of Undernourishment	0	0	0.6277	-0.0564	-0.1188	-0.1186	-0.3103
Proportion Using Safely Managed Drinking Water	0	0	0.1392	-0.0405	-0.1770	0.0131	0.2773
Physicians per 1000 People	-0.1275	0	0	0.1616	-0.0203	-0.0008	0.3169
Nurses and Midwives per 1000 People	0	-0.2068	0	0.1338	0.1737	0.0243	0.2949
Hospital Beds per 1000 People	0	0	0.0125	0.3564	0.2886	0.2492	0.2395
Share of Children Immunized (DTP3)	0	0.3452	0	-0.3747	0.2594	0.0832	0.2387
Total Healthcare Expenditure (% GDP)	0	-0.1044	0	0.0390	0.1430	-0.2069	0.2347
Share of Out-of-Pocket Expenditure on Healthcare	0	0.8137	0	0.1741	0.1722	0.1647	-0.1152
Population	0	-0.0607	0	-0.5870	0.0545	0.6068	-0.0068
GDP per Capita (World Bank)	0	0	0.6559	-0.1452	0.0789	-0.0724	0.2564

Within a refined subspace based on the extracted assumed relevant variables, the DGRC procedure partitions the 105 nations into four distinct clusters, with the memberships detailed in Table 5.4. To provide a clearer interpretation of these groupings, Figure 5.4 displays a geographic visualisation of the clusters on a World Map. The three bivariate projections onto the relevant latent space (LC1$\times$LC2, LC1$\times$LC3, and LC2$\times$LC3) are detailed in the Appendix (Figure A.5). However, the geographic map serves as the principal visualisation, as interpreting fragmented pairwise plots across three dimensions can obscure the global macro-level trends that spatial mapping highlights.

Table 5.4 DGRC Cluster Memberships

Cluster 1	28	Algeria, Azerbaijan, Bangladesh, Cambodia, Cameroon, Egypt, Guatemala, Honduras, Iran, Iraq, Jordan, Kyrgyzstan, Madagascar, Mexico, Morocco, Myanmar, Nepal, Pakistan, Paraguay, Senegal, Sri Lanka, Sudan, Tajikistan, Togo, Tunisia, Turkmenistan, Uzbekistan, Zimbabwe
Cluster 2	22	Angola, Bolivia, Brazil, Burkina Faso, Burundi, China, Colombia, Côte d'Ivoire, Democratic Republic of Congo, Ecuador, Ethiopia, Ghana, Guinea, Indonesia, Laos, Libya, Malawi, Mali, Mozambique, Niger, Turkey, Zambia
Cluster 3	30	Afghanistan, Argentina, Belarus, Benin, Bulgaria, Chad, Chile, Dominican Republic, Greece, Hungary, India, Israel, Kazakhstan, Malaysia, Nicaragua, Nigeria, Peru, Philippines, Poland, Portugal, Russia, Saudi Arabia, Serbia, Sierra Leone, Syria, Tanzania, Uganda, Ukraine, Venezuela, Vietnam
Cluster 4	25	Australia, Austria, Belgium, Canada, Cuba, Czechia, France, Germany, Haiti, Italy, Japan, Kenya, Netherlands, Papua New Guinea, Romania, Rwanda, South Africa, South Korea, Spain, Sweden, Switzerland, Thailand, United Arab Emirates, United Kingdom, United States

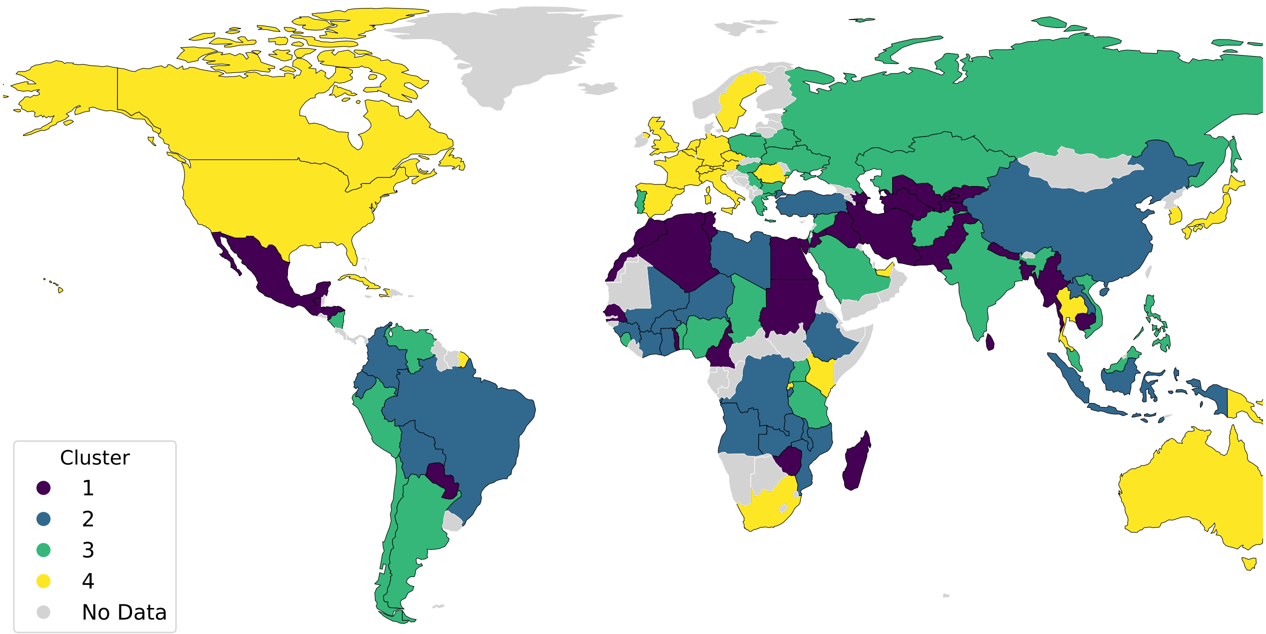

Figure 5.4 Map of DGRC Cluster Memberships

Chapter 6

Conclusion

In this project, we explored simultaneous dimensionality reduction and clustering methodologies, establishing a centralised mathematical framework with consistent notation to evaluate where these procedures perform well and where they can inherently struggle. We demonstrated that standard sequential approaches, such as tandem analysis, are highly vulnerable to the cluster masking problem. While simultaneous methods like Reduced K-means (RKM) and Factorial K-means (FKM) resolve this by coupling the projection and clustering steps into a single objective, they can often produce latent components with overlapping loadings that lack interpretability in terms of the original variables.

To address this, following the ideas of Vichi (2002) and Vichi and Saporta (2009), we introduced disjointness constraints to these models and detailed the primary methodological contribution of this report, the development of a Disjoint Generalised Reduced Clustering (DGRC) procedure. DGRC extends the notion of disjointness onto the GRC model, allowing the construction of an interpretable clustering space while explicitly filtering out disturbing variables via subspace separation. From a theoretical standpoint, DGRC naturally generalises both CDPCA and the DFKM framework, as their respective loss functions are simply specific instances of the DGRC objective where the disturbing dimension $L_D = 0$. To optimise this new objective, we developed a novel loading matrix update loop. Unlike CDPCA, which relies on extensive and computationally expensive singular value decompositions (SVD) at each iteration block, the alternative loop utilises linear scalar projections to efficiently assign variables to disjoint components. Furthermore, it is worth noting that as Vichi and Saporta (2009) mention regarding CDPCA, when the target dimensionality is set to $L = P$, DGRC can effectively act as an alternative to Sparse Principal Component Analysis Zou et al. (2006). It derives L_C sparse and disjoint components through discrete constraints, bypassing the need for lasso or elastic net penalties in the loss function.

The practical advantages of this methodology were demonstrated through our application on the Global Health Indicators dataset. CDPCA struggled with the disproportionate variance of specific metrics, completely isolating the population variable into a single component at the expense of capturing nuanced clustering structures. In contrast, DGRC leverages the notion of subspace separation to filter out these disturbing variables, and thus constructing an interpretable profile of global health capacity without being dominated by isolated high-variance features.

Despite this, the current DGRC framework possesses notable limitations that present clear avenues for future work. Primarily, while the novel subspace loop avoids the need for explicit data whitening, it still inherently relies on RKM-based update logic, which assumes that the variance within the projected clustering-relevant subspace remains locally constant during the updates of the loading matrix $\mathbf{A}$. Future work could focus on constructing a more mathematically rigorous optimisation loop that removes the need for this assumption. Such an improvement must carefully address the tendency to converge toward degenerate low-variance subspaces, which can otherwise cause the FKM objective to deteriorate rapidly and provide uninformative results when data is not priorly whitened, as discussed in Section 4.3. Additionally, the introduction of subspace separation heavily increases the computational burden of model selection. DGRC requires exhaustive cross-validation over an expanded hyperparameter set in order to tune dimensions L_C and L_D, alongside the coefficients ρ_1 and ρ_2. For exceptionally large datasets, this can become a computational burden. Future extensions of this work could explore heuristic techniques or information criteria to more efficiently estimate the optimal subspace dimensions and penalty parameters prior to execution, extending towards a more robust and scalable clustering tool. In addition, the scope of this project did not encompass exceedingly high-dimensional datasets containing more than several hundreds of variables, meaning the evaluation of the behaviour and scalability of DGRC within such high-dimensional regimes presents an avenue for further exploration.

Appendix A

Supplementary Material

A.1 Python Implementations

As discussed in Chapter 5, the simultaneous dimensionality reduction and clustering algorithms (RKM, FKM, CDPCA, and GRC) were initially translated into Python from their original `R`, `C++`, and `C` source codes Prunila and Vichi (2025); Yamamoto (2014). This initial translation was assisted by Claude 4.6 Sonnet; the complete prompt history is preserved in `Claude-prompt-response.txt` wheras the raw Python outputs are stored in `drclust-pyton-imp.py` and `grc-pyton-imp.py`. However these were direct translations and lacked some computational efficiency, so the translations were used as inspiration and the functions were entirely rewritten using a class-based structure `numpy` to handle matrix operations effectively. The final cenrtralised version containing the base code for these clustering procedures is contained within `base.py`, which also contains the original Python implementations for the DFKM loop and the DGRC methodology.

The repository containing the files mentioned above can be accessed at:
https://github.com/mohamedkhussain/simultaneous-clustering.

This repository also includes several Jupyter notebooks which contain the code used to generate the Figures and empirical results throughout this report. Specifically, `tandem_failure.ipynb` contains the code required to produce the comparative scatter plots in Figures 2.1, 3.1, and 4.1. The `synthetic.ipynb` notebook contains the code to generate the synthetic datasets and reproduce the experiments and boxplots detailed in Section 5.1. Finally, `owid_clustering.ipynb` and `owid_data.csv` contain the dataset, clustering procedures, and plotting code for the global health application discussed in Section 5.2.

A.2 Synthetic Simulation Box-plots

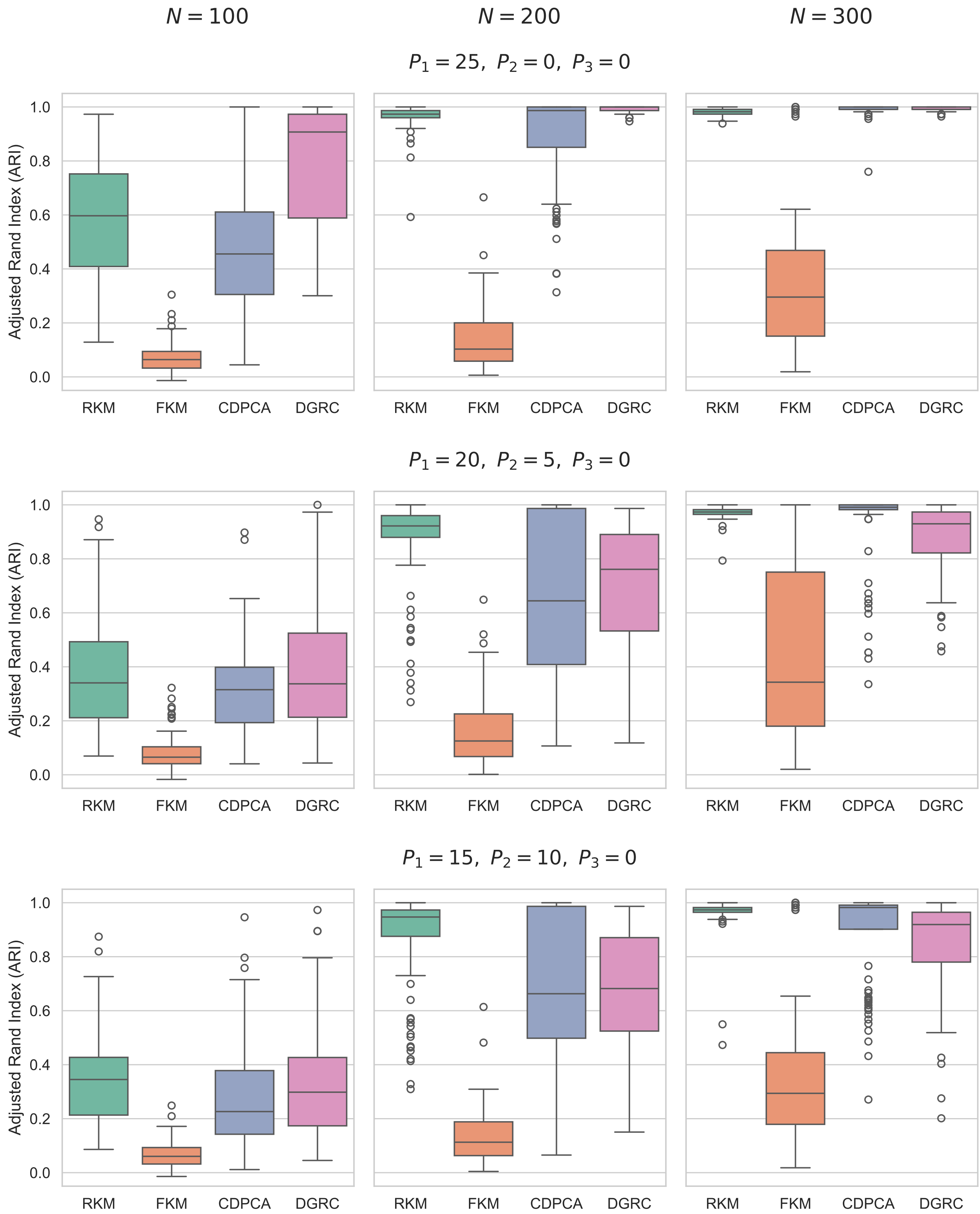

Figure A.1 Boxplots of Adjusted Rand Indices for RKM, FKM, CDPCA and DGRC across values of N, P_1, P_2 with fixed $P_3 = 0$

A.3 Loadings & Projections for Global Health Indicators Dataset

Table A.1 PCA Loadings for Principal Components 1 and 2

Metric	PC1	PC2
Life Expectancy	0.3335	0.1384
Child Mortality	-0.3102	-0.2005
Incidence of Tuberculosis (SDGs)	-0.2264	-0.1866
Share of Adults Who Smoke	0.1520	0.2393
Total Alcohol Consumption per Capita (Litres)	0.2097	-0.3948
Share of Adults Defined as Obese	0.2085	0.2325
Prevalence of Undernourishment	-0.2444	-0.3491
Proportion Using Safely Managed Drinking Water	0.2875	0.0825
Physicians per 1000 People	0.3140	-0.0296
Nurses and Midwives per 1000 People	0.3097	-0.1915
Hospital Beds per 1000 People	0.2416	-0.1801
Share of Children Immunized (DTP3)	0.2107	0.2326
Total Healthcare Expenditure (% GDP)	0.2434	-0.2570
Share of Out-of-Pocket Expenditure on Healthcare	-0.1767	0.4729
Population	-0.0035	0.2640
GDP per Capita (World Bank)	0.3172	-0.1625

Table A.2 RKM loadings

Metric	LC1	LC2
Life Expectancy	0.3369	-0.2088
Child Mortality	-0.3067	0.3518
Incidence of Tuberculosis (SDGs)	-0.2303	0.0565
Share of Adults Who Smoke	0.1640	-0.0917
Total Alcohol Consumption per Capita (Litres)	0.2004	0.4417
Share of Adults Defined as Obese	0.2238	-0.3047
Prevalence of Undernourishment	-0.2552	0.3372
Proportion Using Safely Managed Drinking Water	0.2859	-0.0505
Physicians per 1000 People	0.3316	0.1254
Nurses and Midwives per 1000 People	0.3023	0.1783
Hospital Beds per 1000 People	0.2472	0.2533
Share of Children Immunized (DTP3)	0.2030	-0.2232
Total Healthcare Expenditure (% GDP)	0.2308	0.2715
Share of Out-of-Pocket Expenditure on Healthcare	-0.1429	-0.3171
Population	-0.0089	-0.1983
GDP per Capita (World Bank)	0.3129	0.2041

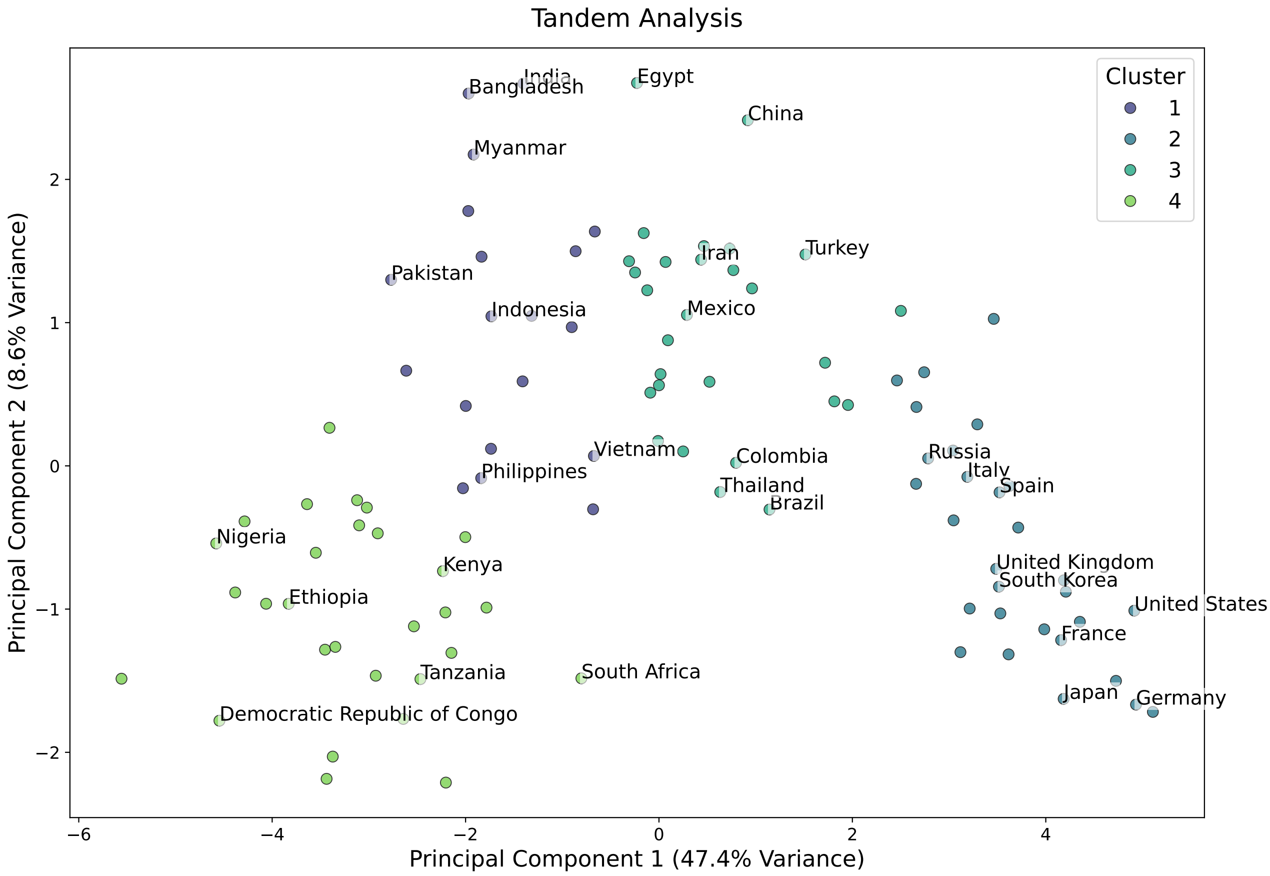

Figure A.2 Tandem Analysis; projection onto first two PCs and K-means clusters

Table A.3 DGRC Loadings with $L_C = 2, L_D = 4$

	Relevant		Disturbing			
Metric	LC1	LC2	LC3	LC4	LC5	LC6
Life Expectancy	0	0.4495	-0.0943	0.1630	0.0994	0.3080
Child Mortality	0	0.6923	-0.0418	0.0305	-0.2495	-0.3541
Incidence of Tuberculosis (SDGs)	0.3082	0	0.2860	-0.2995	0.1817	-0.2386
Share of Adults Who Smoke	0	0.1735	0.5487	-0.0863	0.5317	0.1408
Total Alcohol Consumption per Capita (Litres)	0	0.3680	0.1672	-0.3891	-0.2591	0.1856
Share of Adults Defined as Obese	0.1316	0	0.1438	0.5786	0.0542	0.2065
Prevalence of Undernourishment	0.6826	0	-0.0250	0.0117	-0.0930	-0.2689
Proportion Using Safely Managed Drinking Water	-0.0387	0	0.0955	0.0513	0.0776	0.2900
Physicians per 1000 People	0.1770	0	0.0924	0.0518	0.0321	0.3081
Nurses and Midwives per 1000 People	0	0.1885	-0.0001	-0.1265	-0.1367	0.2976
Hospital Beds per 1000 People	0.1085	0	0.1254	-0.3455	0.0272	0.2369
Share of Children Immunized (DTP3)	0	0.1047	-0.4555	0.0192	0.1442	0.2061
Total Healthcare Expenditure (% GDP)	-0.0799	0	-0.0190	-0.0486	-0.3215	0.2461
Share of Out-of-Pocket Expenditure on Healthcare	0	0.3211	-0.0197	0.3339	0.3471	-0.1970
Population	0	0.0596	-0.5275	-0.3617	0.5097	-0.0076
GDP per Capita (World Bank)	0.6089	0	-0.1933	0.0569	-0.0508	0.2965

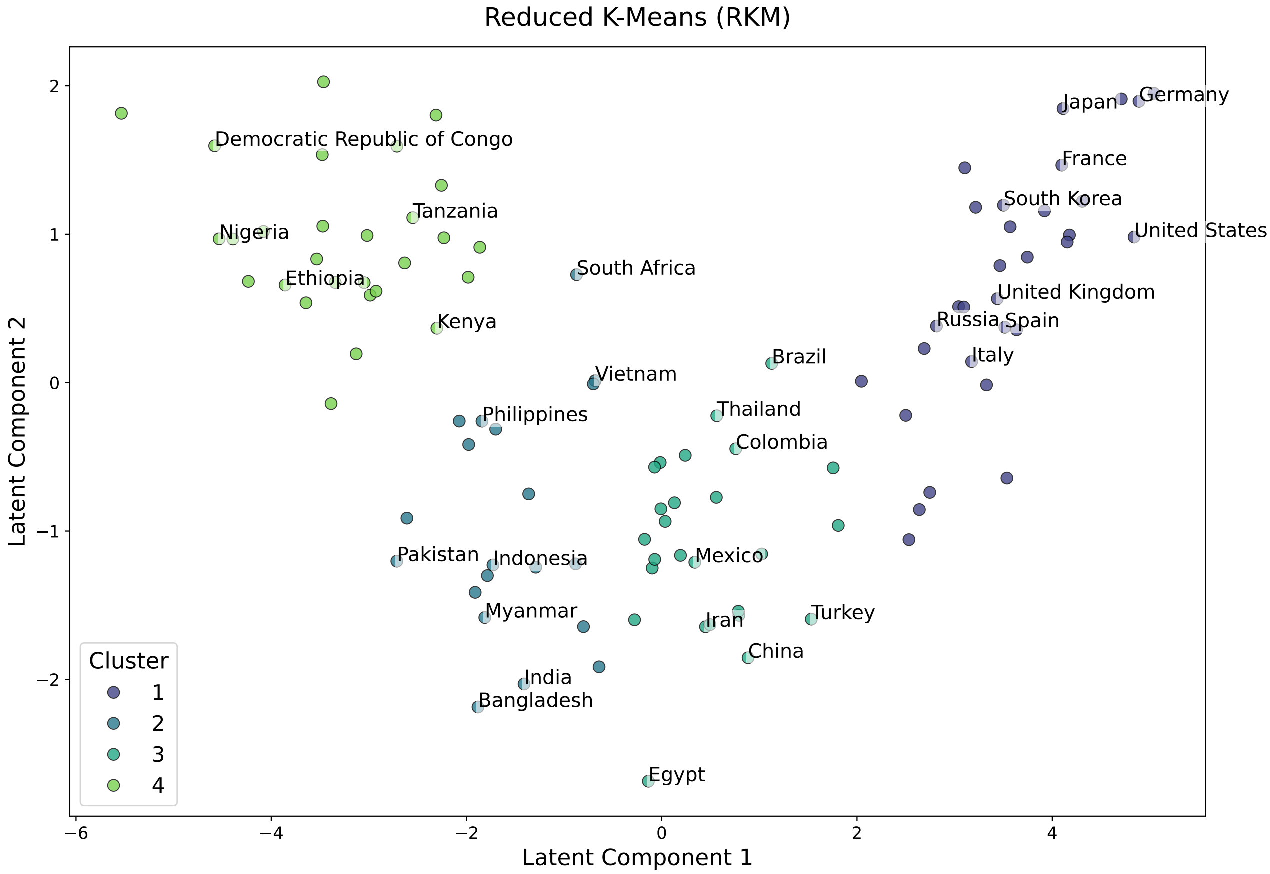

Figure A.3 RKM projections onto latent components

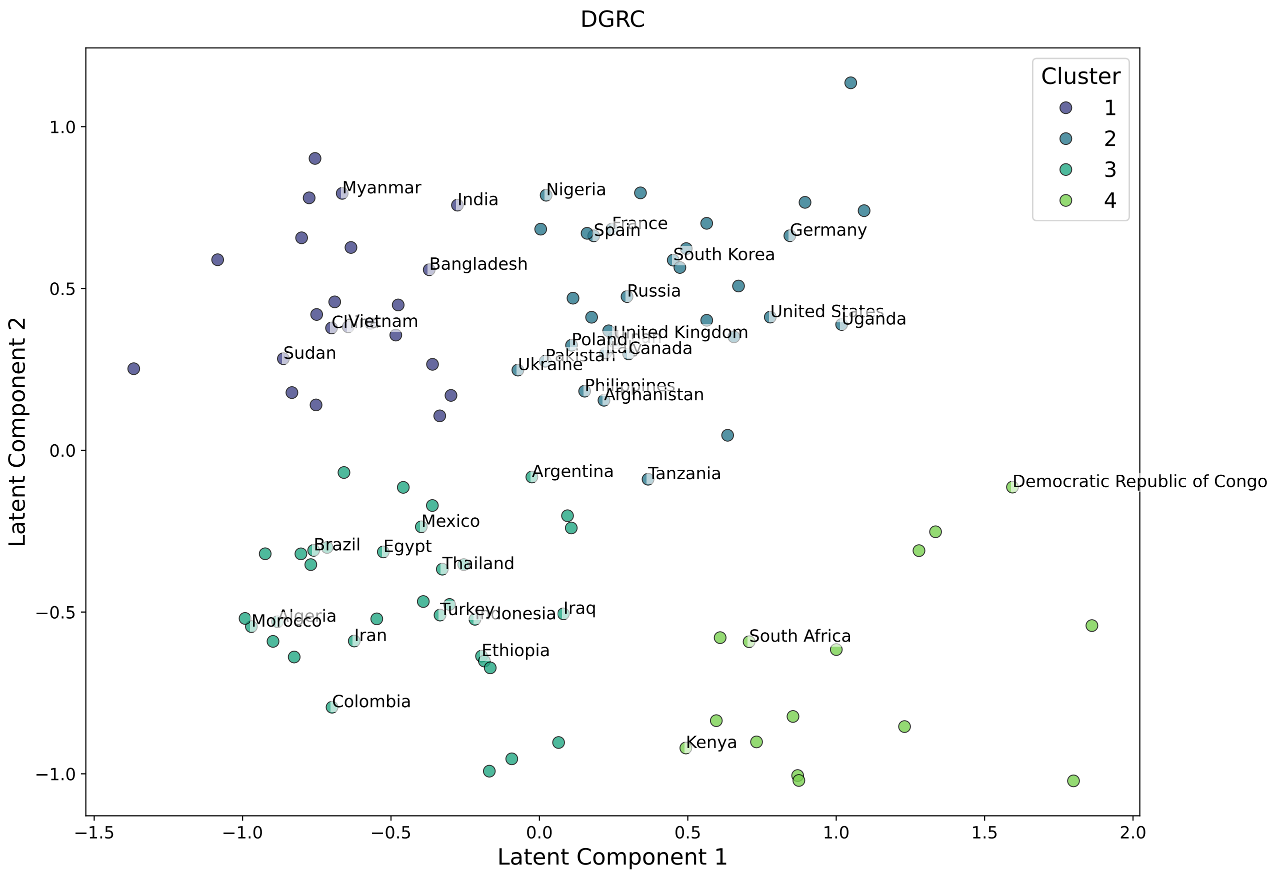

Figure A.4 DGRC projections onto latent component with $L_C = 2$

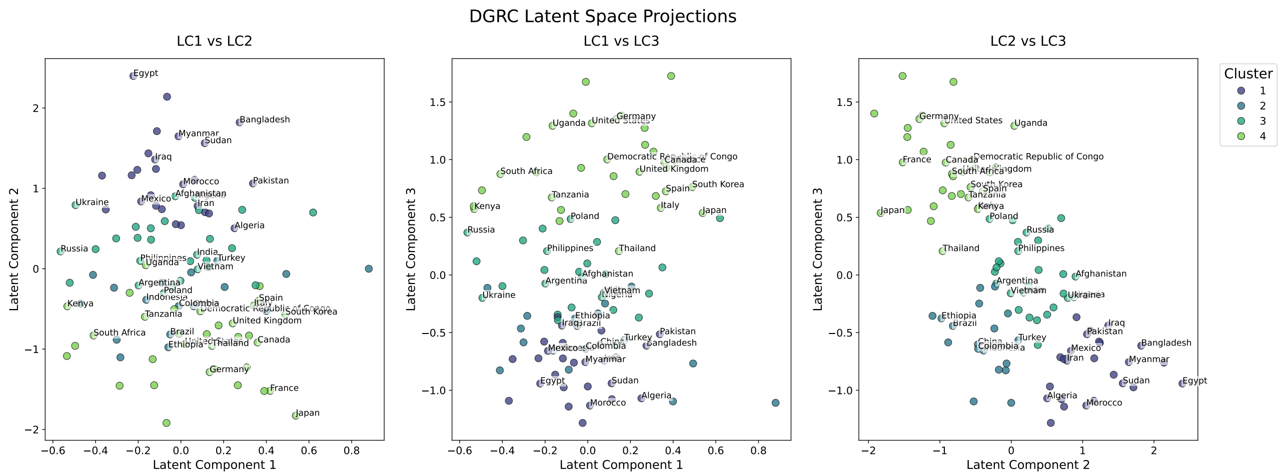

Figure A.5 DGRC projections onto latent components with $L_C = 3$

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